

Pathway Analysis Report

This report contains the pathway analysis results for the submitted sample ". Analysis was performed against Reactome version 86 on 01/10/2023. The web link to these results is:

<https://reactome.org/PathwayBrowser/#/ANALYSIS=MjAyMzEwMDEwNTU4NDRfMjE5NTY%3D>

Please keep in mind that analysis results are temporarily stored on our server. The storage period depends on usage of the service but is at least 7 days. As a result, please note that this URL is only valid for a limited time period and it might have expired.

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1. Introduction

Reactome is a curated database of pathways and reactions in human biology. Reactions can be considered as pathway 'steps'. Reactome defines a 'reaction' as any event in biology that changes the state of a biological molecule. Binding, activation, translocation, degradation and classical biochemical events involving a catalyst are all reactions. Information in the database is authored by expert biologists, entered and maintained by Reactome's team of curators and editorial staff. Reactome content frequently cross-references other resources e.g. NCBI, Ensembl, UniProt, KEGG (Gene and Compound), ChEBI, PubMed and GO. Orthologous reactions inferred from annotation for Homo sapiens are available for 14 non-human species including mouse, rat, chicken, puffer fish, worm, fly and yeast. Pathways are represented by simple diagrams following an SBGN-like format.

Reactome's annotated data describe reactions possible if all annotated proteins and small molecules were present and active simultaneously in a cell. By overlaying an experimental dataset on these annotations, a user can perform a pathway over-representation analysis. By overlaying quantitative expression data or time series, a user can visualize the extent of change in affected pathways and its progression. A binomial test is used to calculate the probability shown for each result, and the p-values are corrected for the multiple testing (Benjamini-Hochberg procedure) that arises from evaluating the submitted list of identifiers against every pathway.

To learn more about our Pathway Analysis, please have a look at our relevant publications:

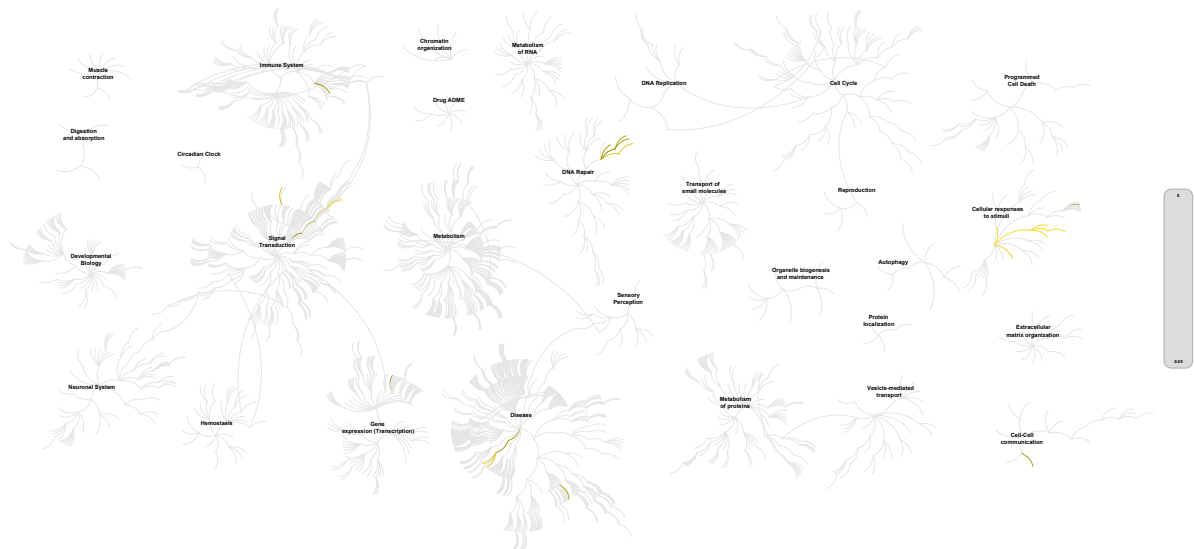
Fabregat A, Sidiropoulos K, Garapati P, Gillespie M, Hausmann K, Haw R, ... D'Eustachio P (2016). The reactome pathway knowledgebase. *Nucleic Acids Research*, 44(D1), D481–D487. <https://doi.org/10.1093/nar/gkv1351>. 

Fabregat A, Sidiropoulos K, Viteri G, Forner O, Marin-Garcia P, Arnau V, ... Hermjakob H (2017). Reactome pathway analysis: a high-performance in-memory approach. *BMC Bioinformatics*, 18. 

2. Properties

- This is an **overrepresentation** analysis: A statistical (hypergeometric distribution) test that determines whether certain Reactome pathways are over-represented (enriched) in the submitted data. It answers the question 'Does my list contain more proteins for pathway X than would be expected by chance?' This test produces a probability score, which is corrected for false discovery rate using the Benjamani-Hochberg method. [↗](#)
- 5 out of 6 identifiers in the sample were found in Reactome, where 366 pathways were hit by at least one of them.
- All non-human identifiers have been converted to their human equivalent. [↗](#)
- IntAct interactors were included to increase the analysis background. This greatly increases the size of Reactome pathways, which maximises the chances of matching your submitted identifiers to the expanded pathway, but will include interactors that have not undergone manual curation by Reactome and may include interactors that have no biological significance, or unexplained relevance.
- This report is filtered to show only results for species 'Homo sapiens' and resource 'all resources'.
- The unique ID for this analysis (token) is MjAyMzEwMDEwNTU4NDRfMjE5NTY%3D. This ID is valid for at least 7 days in Reactome's server. Use it to access Reactome services with your data.

3. Genome-wide overview



This figure shows a genome-wide overview of the results of your pathway analysis. Reactome pathways are arranged in a hierarchy. The center of each of the circular "bursts" is the root of one top-level pathway, for example "DNA Repair". Each step away from the center represents the next level lower in the pathway hierarchy. The color code denotes over-representation of that pathway in your input dataset. Light grey signifies pathways which are not significantly over-represented.

4. Most significant pathways

The following table shows the 25 most relevant pathways sorted by p-value.

Pathway name	Entities				Reactions	
	found	ratio	p-value	FDR*	found	ratio
Attenuation phase	7 / 47	0.002	1.23e-14	4.93e-12	3 / 5	3.45e-04
HSF1-dependent transactivation	7 / 59	0.003	6.02e-14	1.20e-11	4 / 8	5.52e-04
Regulation of HSF1-mediated heat shock response	9 / 260	0.011	5.07e-13	6.74e-11	7 / 14	9.67e-04
Cellular response to heat stress	9 / 305	0.013	2.09e-12	2.09e-10	12 / 29	0.002
HSF1 activation	6 / 98	0.004	2.38e-10	1.90e-08	1 / 7	4.83e-04
Cellular responses to stress	13 / 3,758	0.165	9.83e-05	0.006	62 / 494	0.034
Cellular responses to stimuli	13 / 3,800	0.167	1.09e-04	0.006	62 / 525	0.036
NGF-stimulated transcription	3 / 384	0.017	0.004	0.222	24 / 37	0.003
Nuclear Events (kinase and transcription factor activation)	3 / 430	0.019	0.006	0.247	24 / 48	0.003
HSP90 chaperone cycle for steroid hormone receptors (SHR) in the presence of ligand	2 / 134	0.006	0.006	0.247	8 / 22	0.002
Evasion of Oxidative Stress Induced Senescence Due to p14ARF Defects	1 / 14	6.17e-04	0.012	0.276	1 / 1	6.91e-05
Evasion of Oncogene Induced Senescence Due to p14ARF Defects	1 / 14	6.17e-04	0.012	0.276	1 / 1	6.91e-05
Abasic sugar-phosphate removal via the single-nucleotide replacement pathway	1 / 14	6.17e-04	0.012	0.276	1 / 2	1.38e-04
Resolution of AP sites via the single-nucleotide replacement pathway	1 / 17	7.49e-04	0.015	0.276	1 / 6	4.14e-04
Estrogen-dependent nuclear events downstream of ESR-membrane signaling	2 / 241	0.011	0.019	0.276	2 / 12	8.29e-04
Signaling by NTRK1 (TRKA)	3 / 703	0.031	0.023	0.276	24 / 102	0.007
Viral RNP Complexes in the Host Cell Nucleus	1 / 34	0.001	0.03	0.276	1 / 3	2.07e-04
PCNA-Dependent Long Patch Base Excision Repair	1 / 39	0.002	0.034	0.276	1 / 6	4.14e-04
POLB-Dependent Long Patch Base Excision Repair	1 / 39	0.002	0.034	0.276	1 / 10	6.91e-04
Displacement of DNA glycosylase by APEX1	1 / 43	0.002	0.037	0.276	8 / 8	5.52e-04
Diseases of cellular response to stress	1 / 43	0.002	0.037	0.276	1 / 3	2.07e-04
Diseases of Cellular Senescence	1 / 43	0.002	0.037	0.276	1 / 3	2.07e-04

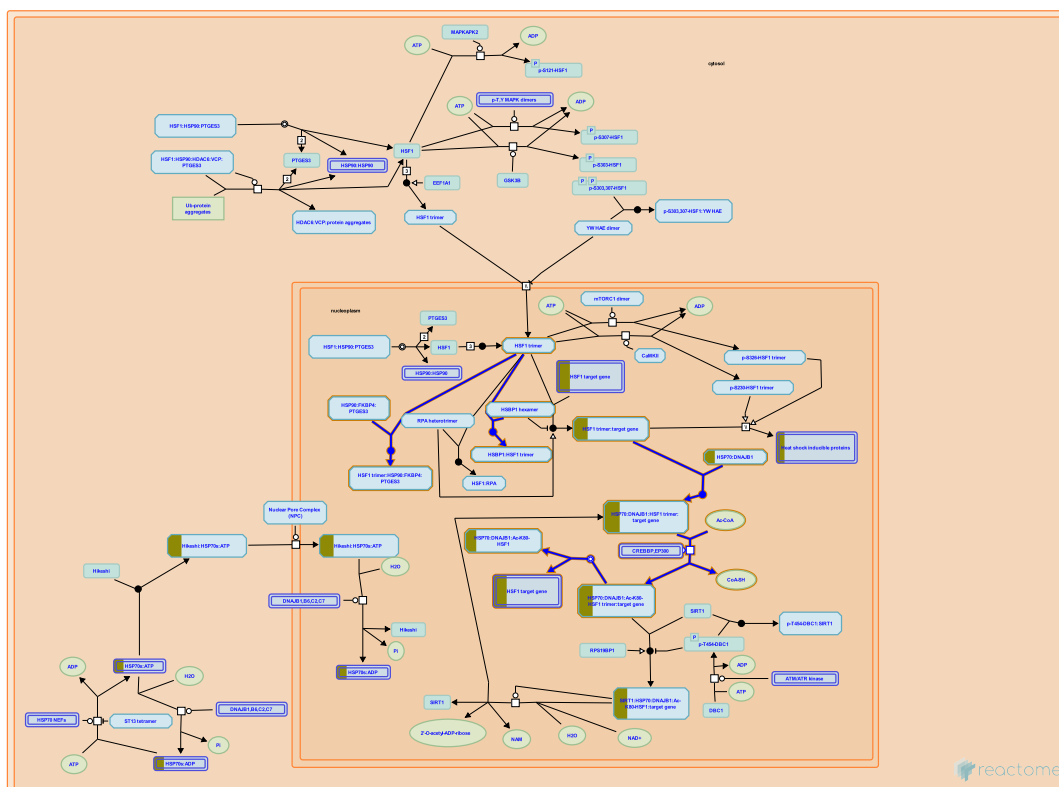
Pathway name	Entities				Reactions	
	found	ratio	p-value	FDR*	found	ratio
Signal regulatory protein family interactions	1 / 45	0.002	0.039	0.276	1 / 10	6.91e-04
Interleukin-4 and Interleukin-13 signaling	2 / 363	0.016	0.04	0.276	1 / 47	0.003
NFE2L2 regulating ER-stress associated genes	1 / 51	0.002	0.044	0.276	1 / 2	1.38e-04

* False Discovery Rate

5. Pathways details

For every pathway of the most significant pathways, we present its diagram, as well as a short summary, its bibliography and the list of inputs found in it.

1. Attenuation phase (R-HSA-3371568)



Attenuation of the heat shock transcriptional response occurs during continuous exposure to intermediate heat shock conditions or upon recovery from stress (Abravaya et al. 1991). The attenuation phase of HSF1 cycle involves the transcriptional silencing of HSF1 bound to HSE, the release of HSF1 trimers from HSE and dissociation of HSF1 trimers to monomers. HSF1-driven heat stress associated transcription was shown to depend on inducible and reversible acetylation of HSF1 at Lys80, which negatively regulates DNA binding activity of HSF1 (Westerheide SD et al. 2009). In addition, the attenuation of HSF1 activation takes place when enough HSP70/HSP40 is produced to saturate exposed hydrophobic regions of proteins damaged as a result of heat exposure. The excess HSP70/HSP40 binds to HSF1 trimer, which leads to its dissociation from the promoter and conversion to the inactive monomeric form (Abravaya et al. 1991; Shi Y et al. 1998). Interaction of HSP70 with the transcriptional corepressor repressor element 1-silencing transcription factor corepressor (CoREST) assists in terminating heat-shock response (Gomez AV et al. 2008). HSF1 DNA-binding and transactivation activity were also inhibited upon interaction of HSF1-binding protein (HSBP1) with active trimeric HSF1 (Satval SH et al. 1998).

References

Morimoto RI, Abravaya K & Phillips B (1991). Attenuation of the heat shock response in HeLa cells is mediated by the release of bound heat shock transcription factor and is modulated by changes in growth and in heat shock temperatures. *Genes Dev.*, 5, 2117-27. [🔗](#)

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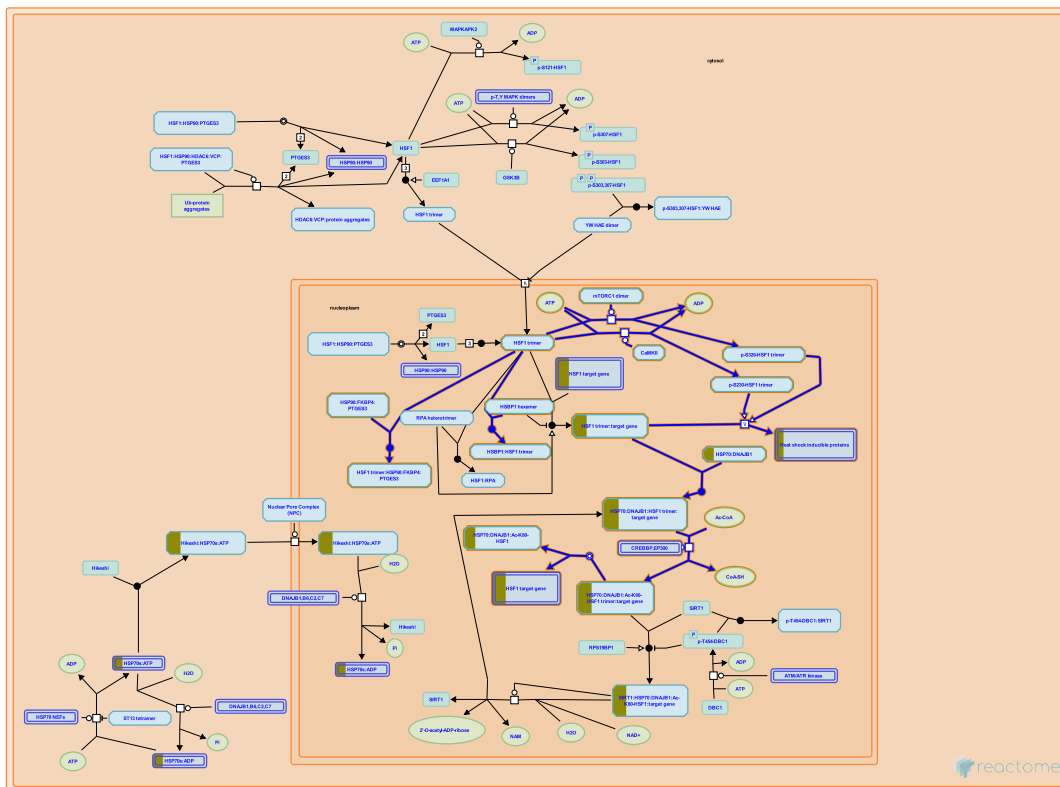
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2013-10-29	Authored	Shamovsky V
2014-02-17	Edited	Shamovsky V
2014-02-17	Reviewed	Pani B
2023-08-26	Modified	Wright A

2 submitted entities found in this pathway, mapping to 7 Reactome entities

Input	UniProt Id
HSPA1A	P0DMV8

Input	Ensembl Id	Input	Ensembl Id
HSPA1A	ENSG00000204389, ENSG00000215328, ENSG00000234475, ENSG00000235941, ENSG00000237724	HSPA6	ENSG00000173110

2. HSF1-dependent transactivation (R-HSA-3371571)



Acquisition of DNA binding activity by HSF1 is necessary but insufficient for transcriptional activation (Cotto JJ et al. 1996; Trinklein ND et al. 2004). In addition to having a sequence-specific DNA binding domain, HSF1 contains a C-terminal region which is involved in activating the transcription of the target genes (Green M et al. 1995). However, the transactivating ability of the transactivation domain itself is not stress sensitive. Rather, it's controlled by a regulatory domain of HSF1 (amino acids 221-310), which represses the transactivating ability under normal physiological conditions (Green M et al. 1995; Zuo J et al. 1995; Newton EM et al. 1996). The HSF1 transactivation domain can be divided into two distinct regions, activation domain 1 (AD1) and activation domain 2 (AD2) (Brown SA et al. 1998). AD1 and AD2 each contain residues that are important for both transcriptional initiation and elongation. Mutations in acidic residues in both AD1 and AD2 preferentially affect the ability of HSF1 to stimulate transcriptional initiation, while mutations in phenylalanine residues preferentially affect stimulation of elongation (Brown SA et al. 1998).

Activation of the DNA-bound but transcriptionally incompetent HSF1 is thought to occur upon stress induced HSF1 phosphorylation at several serine residues (Ding XZ et al. 1997; Holmberg CI et al. 2001; Guettouche T et al. 2005). In cells exposed to heat, acquisition of HSE DNA-binding activity was observed to precede phosphorylation of HSF1 (Cotto JJ et al. 1996; Kline MP & Morimoto RI 1997). While there is a sufficient evidence to suggest that phosphorylation of HSF1 is essential to modulate HSF1 transactivating capacity, mechanisms behind stress stimuli and kinases/phosphatases involved have not been clearly established.

References

Rungger D, Voellmy R & Zuo J (1995). Multiple layers of regulation of human heat shock transcription factor 1. *Mol. Cell. Biol.*, 15, 4319-30. [🔗](#)

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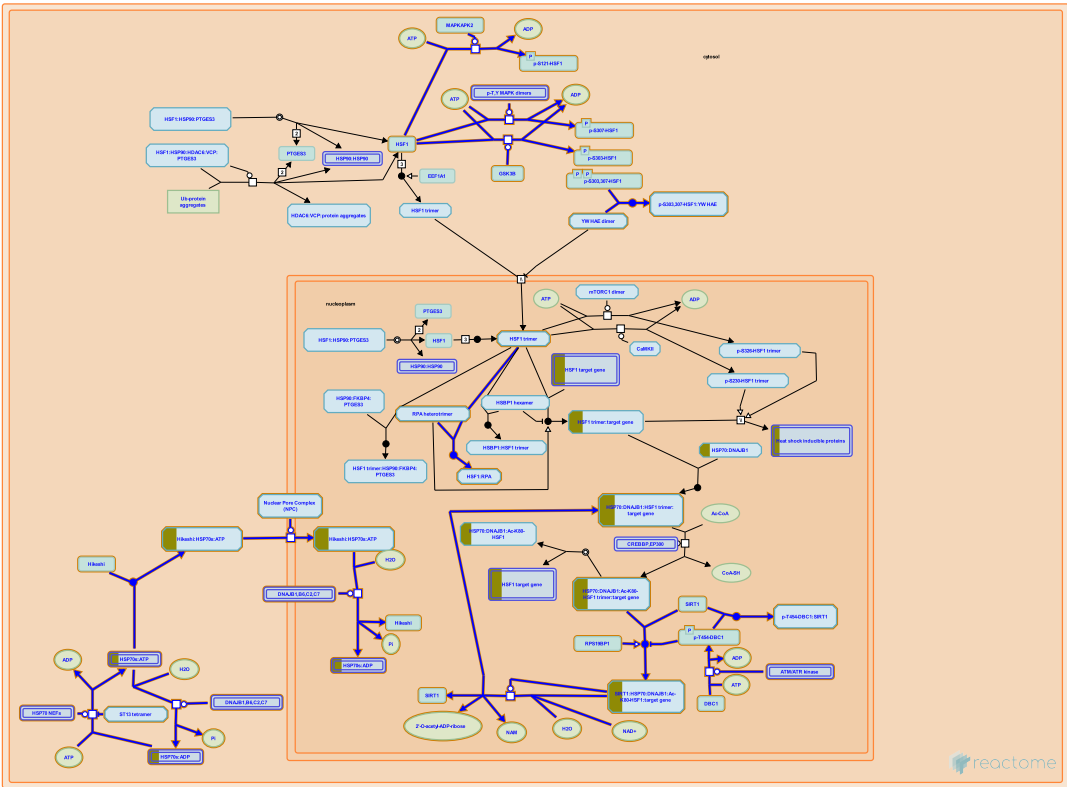
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2013-10-29	Authored	Shamovsky V
2014-02-17	Edited	Shamovsky V
2014-02-17	Reviewed	Pani B
2023-08-26	Modified	Wright A

2 submitted entities found in this pathway, mapping to 7 Reactome entities

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Input	Ensembl Id	Input	Ensembl Id
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3. Regulation of HSF1-mediated heat shock response (R-HSA-3371453)



The ability of HSF1 to respond to cellular stresses is under negative regulation by chaperones, modulation of nucleocytoplasmic shuttling, post-translational modifications and transition from monomeric to trimeric state.

References

Rungger D, Voellmy R & Zuo J (1995). Multiple layers of regulation of human heat shock transcription factor 1. Mol. Cell. Biol., 15, 4319-30.

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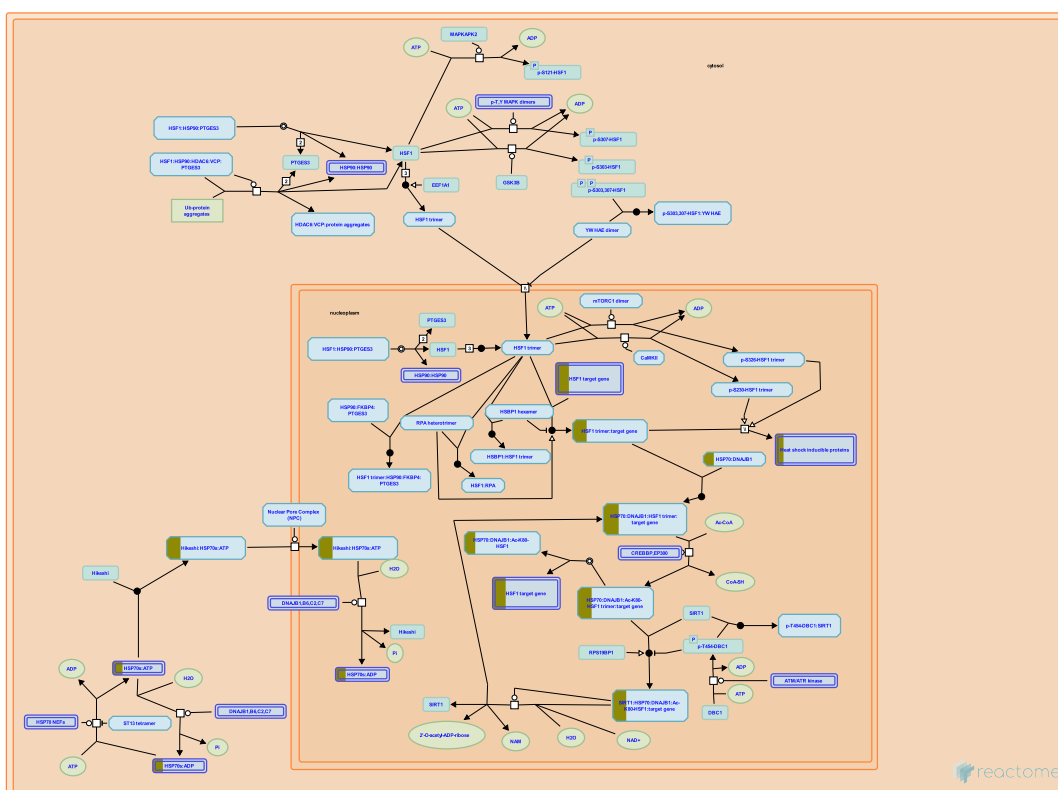
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2014-02-17	Edited	Shamovsky V
2014-02-17	Reviewed	Pani B
2023-08-26	Modified	Wright A

2 submitted entities found in this pathway, mapping to 9 Reactome entities

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Input	Ensembl Id	Input	Ensembl Id
HSPA1A	ENSG00000204389, ENSG00000215328, ENSG00000234475, ENSG00000235941, ENSG00000237724	HSPA6	ENSG00000173110

4. Cellular response to heat stress (R-HSA-3371556)



In response to exposure to elevated temperature and certain other proteotoxic stimuli (e.g., hypoxia, free radicals) cells activate a number of cytoprotective mechanisms known collectively as "heat shock response". Major aspects of the heat shock response (HSR) are evolutionarily conserved events that allow cells to recover from protein damage induced by stress (Liu XD et al. 1997; Voellmy R & Boellmann F 2007; Shamovsky I & Nudler E 2008; Anckar J & Sistonen L 2011). The main hallmark of HSR is the dramatic alteration of the gene expression pattern. A diverse group of protein genes is induced by the exposure to temperatures 3-5 degrees higher than physiological. Functionally, most of these genes are molecular chaperones that ensure proper protein folding and quality control to maintain cell proteostasis.

At the same time, heat shock-induced phosphorylation of translation initiation factor eIF2 α leads to the shutdown of the nascent polypeptide synthesis reducing the burden on the chaperone system that has to deal with the increased amount of misfolded and thermally denatured proteins (Duncan RF & Hershey JWB 1989; Sarkar A et al. 2002; Spriggs KA et al. 2010).

The induction of HS gene expression primarily occurs at the level of transcription and is mediated by heat shock transcription factor HSF1 (Sarge KD et al. 1993; Baler R et al. 1993). Human cells express five members of HSF protein family: HSF1, HSF2, HSF4, HSFX and HSFY. HSF1 is the master regulator of the heat inducible gene expression (Zuo J et al. 1995; Akerfelt M et al. 2010). HSF2 is activated in response to certain developmental stimuli in addition to being co-activated with HSF1 to provide promoter-specific fine-tuning of the HS response by forming heterotrimeric complexes with HSF1 (Ostling P et al. 2007; Sandqvist A et al. 2009). HSF4 lacks the transcription activation domain and acts as a repressor of certain genes during HS (Nakai A et al. 1997; Tanabe M et al. 1999; Kim SA et al. 2012). Two additional family members HSFX and HSFY, which are located on the X and Y chromosomes respectively, remain to be characterized (Bhowmick BK et al. 2006; Shinka T et al. 2004; Kichine E et al. 2012).

Under normal conditions HSF1 is present in both cytoplasm and nucleus in the form of an inactive monomer. The monomeric state of HSF1 is maintained by an intricate network of protein-protein interactions that include the association with HSP90 multichaperone complex, HSP70/HSP40 chaperone machinery, as well as intramolecular interaction of two conserved hydrophobic repeat regions. Monomeric HSF1 is constitutively phosphorylated on Ser303 and Ser 307 by (Zou J et al. 1998; Knauf U et al. 1996; Kline MP & Morimoto RI 1997; Guettouche T et al. 2005). This phosphorylation plays an essential role in ensuring cytoplasmic localization of at least a subpopulation of HSF1 molecules under normal conditions (Wang X et al. 2004).

Exposure to heat and other proteotoxic stimuli results in the release of HSF1 from the inhibitory complex with chaperones and its subsequent trimerization, which is promoted by its interaction with translation elongation factor eEF1A1 (Baler R et al. 1993; Shamovsky I et al. 2006; Herbomel G et al 2013). The trimerization is believed to involve intermolecular interaction between hydrophobic repeats 1-3 leading to the formation of a triple coil structure. Additional stabilization of the HSF1 trimer is provided by the formation of intermolecular S-S bonds between Cys residues in the DNA binding domain (Lu M et al.2008). Trimeric HSF1 is predominantly localized in the nucleus where it binds the specific sequence in the promoter of hsp genes (Sarge KD et al. 1993; Wang Y and Morgan WD 1994). The binding sequence for HSF1 (HSE, heat shock element) contains series of inverted repeats nGAAn in head-to-tail orientation, with at least three elements being required for the high affinity binding. Binding of the HSF1 trimer to the promoter is not sufficient to induce transcription of the gene (Cotto J et al. 1996). In order to do so, HSF1 needs to undergo inducible phosphorylation on specific Ser residues such as Ser230, Ser326. This phosphorylated form of HSF1 trimer is capable of increasing the promoter initiation rate. HSF1 bound to DNA promotes recruiting components of the transcription mediator complex and relieving promoter-proximal pause of RNA polymerase II through its interaction with TFIIF transcription factor (Yuan CX & Gurley WB 2000).

HSF1 activation is regulated in a precise and tight manner at multiple levels (Zuo J et al. 1995; Cotto J et al. 1996). This allows fast and robust activation of HS response to minimize proteotoxic effects of the stress. The exact set of HSF1 inducible genes is probably cell type specific. Moreover, cells in different pathophysiological states will display different but overlapping profile of HS inducible genes.

References

- Voellmy R, Baler R & Dahl G (1993). Activation of human heat shock genes is accompanied by oligomerization, modification, and rapid translocation of heat shock transcription factor HSF1. *Mol. Cell. Biol.*, 13, 2486-96. [↗](#)
- Murphy SP, Morimoto RI & Sarge KD (1993). Activation of heat shock gene transcription by heat shock factor 1 involves oligomerization, acquisition of DNA-binding activity, and nuclear localization and can occur in the absence of stress. *Mol. Cell. Biol.*, 13, 1392-407. [↗](#)

Edit history

Date	Action	Author
2013-05-13	Created	Shamovsky V
2013-10-29	Authored	Shamovsky V
2014-02-17	Edited	Shamovsky V
2014-02-17	Reviewed	Pani B

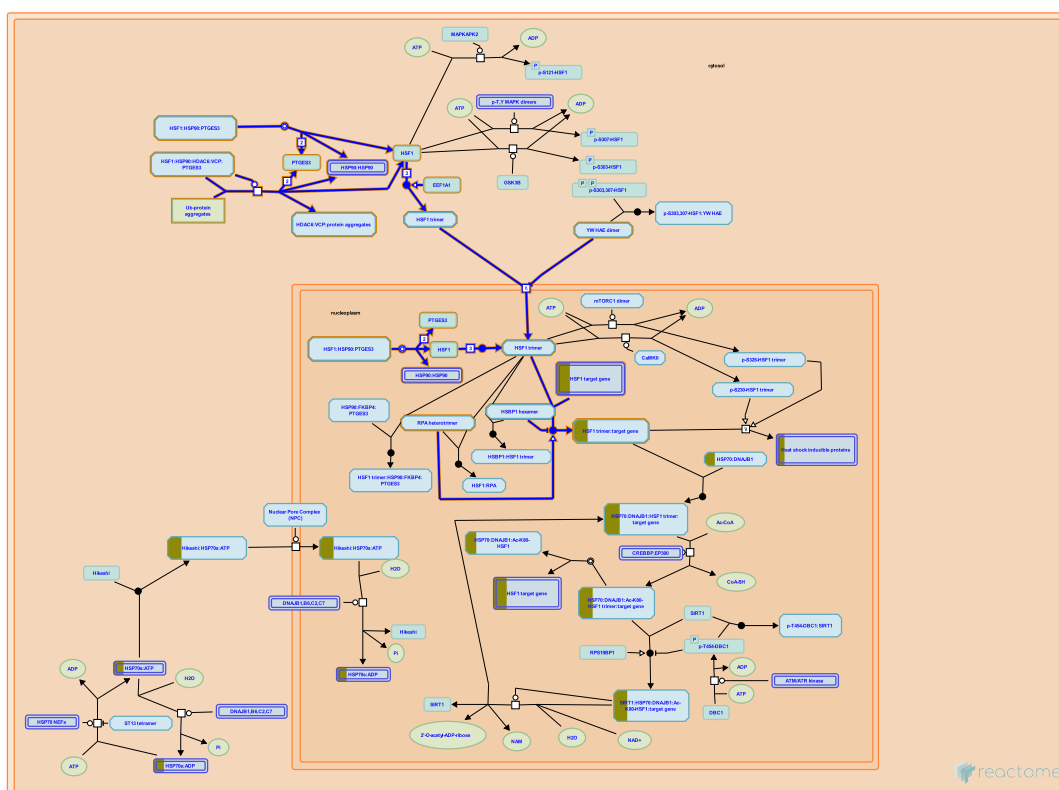
Date	Action	Author
2023-08-26	Modified	Wright A

2 submitted entities found in this pathway, mapping to 9 Reactome entities

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HSPA1A	P0DMV8	HSPA6	P17066, P48741

Input	Ensembl Id	Input	Ensembl Id
HSPA1A	ENSG00000204389, ENSG00000215328, ENSG00000234475, ENSG00000235941, ENSG00000237724	HSPA6	ENSG00000173110

5. HSF1 activation ([R-HSA-3371511](#))



Heat shock factor 1 (HSF1) is a transcription factor that activates gene expression in response to a variety of stresses, including heat shock, oxidative stress, as well as inflammation and infection (Shamovsky I and Nudler E 2008; Akerfelt et al. 2010; Bjork and Sistonen 2010; Anckar and Sistonen 2011).

HSF1 is constitutively present in the cell. In the absence of stress HSF1 is found in both the cytoplasm and the nucleus as an inactive monomer (Sarge KD et al. 1993; Mercier PA et al. 1999; Vujanac M et al. 2005). A physical or chemical proteotoxic stress rapidly induces HSF1 activation, which occurs through a multi-step process, involving HSF1 monomer-to-homotrimer transition, nuclear accumulation, and binding to a promoter element, called the heat shock element (HSE), which leads to the increase in the stress-inducible gene expression (Sarge KD et al. 1993; Baler R et al. 1998; Sonna LA et al. 2002; Shamovsky I and Nudler E 2008; Sakurai H and Enoki Y 2010; Herbolme G et al. 2013). Depending on the type of stress stimulus, the multiple events associated with HSF1 activation might be affected differently (Holmberg CI et al 2000; Bjork and Sistonen 2010).

References

- Morimoto RI, Kline M & Cotto JJ (1996). Activation of heat shock factor 1 DNA binding precedes stress-induced serine phosphorylation. Evidence for a multistep pathway of regulation. *J. Biol. Chem.*, 271, 3355-8. [🔗](#)
- Rungger D, Voellmy R & Zuo J (1995). Multiple layers of regulation of human heat shock transcription factor 1. *Mol. Cell. Biol.*, 15, 4319-30. [🔗](#)

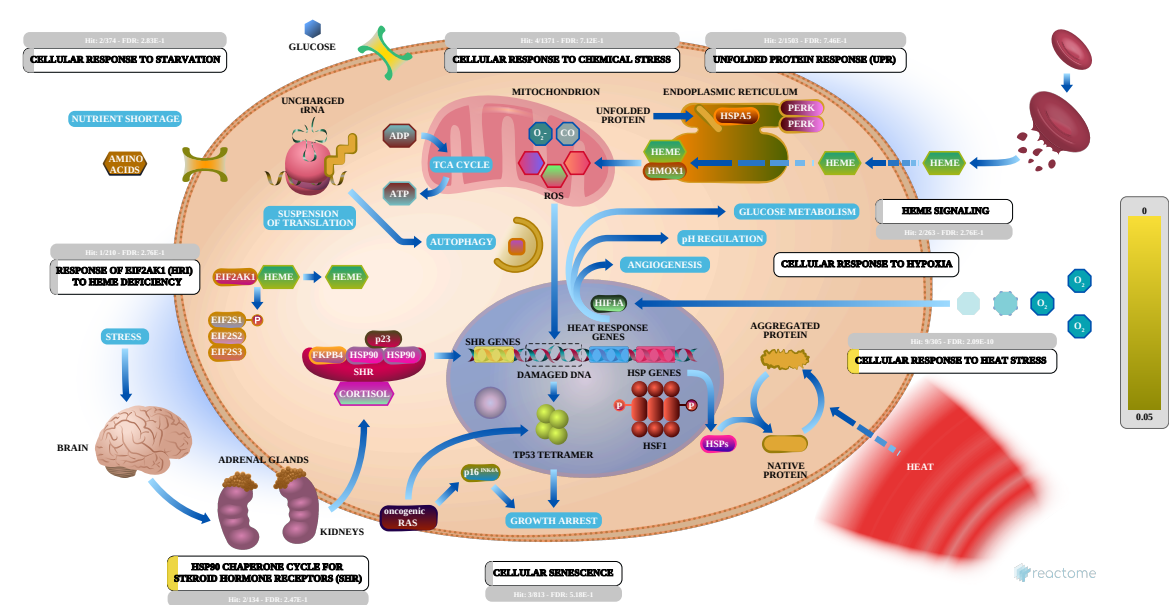
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2013-05-13	Created	Shamovsky V
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2014-02-17	Edited	Shamovsky V
2014-02-17	Reviewed	Pani B
2023-08-26	Modified	Wright A

2 submitted entities found in this pathway, mapping to 6 Reactome entities

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6. Cellular responses to stress (R-HSA-2262752)



Cells are subject to external molecular and physical stresses such as foreign molecules that perturb metabolic or signaling processes, and changes in temperature or pH. Cells are also subject to internal molecular stresses such as production of reactive metabolic byproducts. The ability of cells and tissues to modulate molecular processes in response to such stresses is essential to the maintenance of tissue homeostasis (Kultz 2005). Specific stress-related processes annotated here are **cellular response to hypoxia**, **cellular response to heat stress**, **cellular senescence**, **HSP90 chaperone cycle for steroid hormone receptors (SHR) in the presence of ligand**, **response of EIF2AK1 (HRI) to heme deficiency**, **heme signaling**, **cellular response to chemical stress**, **cellular response to starvation**, and **unfolded protein response**.

References

Kultz D (2005). Molecular and evolutionary basis of the cellular stress response. *Annu. Rev. Physiol.* , 67, 225-57. [🔗](#)

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Date	Action	Author
2012-05-20	Edited	Matthews L
2012-05-20	Reviewed	D'Eustachio P
2012-05-20	Authored	Matthews L
2012-05-20	Created	Matthews L
2023-08-26	Modified	Wright A

4 submitted entities found in this pathway, mapping to 11 Reactome entities

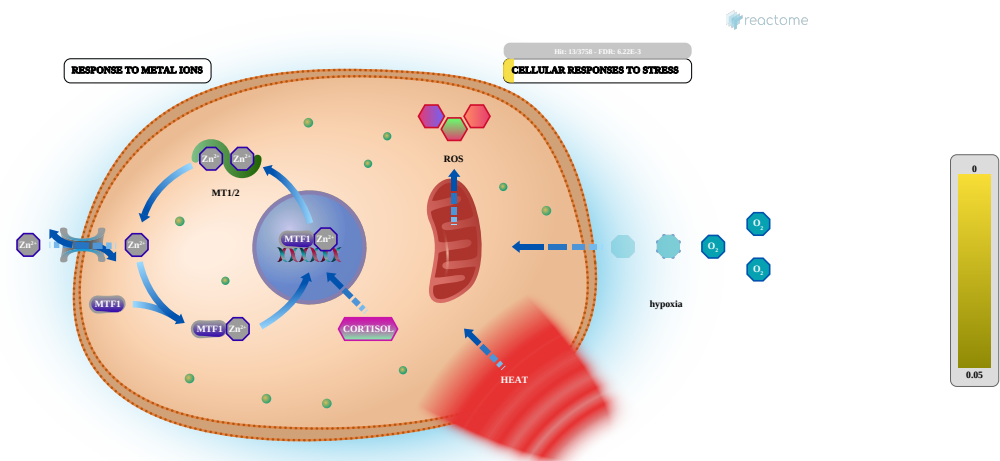
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Interactors found in this pathway (3)

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EGR1	P18146	Q8N726	FOS	Q6FG41, P01100	P05412, P35638, Q96L91, P18848, P15336, P54646, P01100, Q8N6T3
HSPA1A	P0DMV8	Q13501, Q9NUX5, Q9GZQ8			

7. Cellular responses to stimuli (R-HSA-8953897)



Individual cells detect and respond to diverse external molecular and physical signals. Appropriate responses to these signals are essential for normal development, maintenance of homeostasis in mature tissues, and effective defensive responses to potentially noxious agents (Kultz 2005). It is convenient, if somewhat arbitrary, to distinguish responses to signals involved in development and homeostasis from ones involved in stress responses, and that classification is followed here, with **macroautophagy** and **responses to metal ions** classified as responses to normal external stimuli, while responses to hypoxia, reactive oxygen species, and heat, and the process of cellular senescence are classified as **stress responses**. Signaling cascades are integral components of all of these response mechanisms but because of their number and diversity, they are grouped in a separate signal transduction superpathway in Reactome.

References

Kultz D (2005). Molecular and evolutionary basis of the cellular stress response. *Annu. Rev. Physiol.*, 67, 225-57. [🔗](#)

Edit history

Date	Action	Author
2012-05-20	Reviewed	D'Eustachio P
2015-05-13	Reviewed	Tooze SA
2015-09-03	Reviewed	Klionsky DJ
2015-09-19	Reviewed	Atrian S
2016-12-30	Edited	D'Eustachio P
2016-12-30	Authored	D'Eustachio P
2016-12-30	Created	D'Eustachio P
2023-08-26	Modified	Wright A

4 submitted entities found in this pathway, mapping to 11 Reactome entities

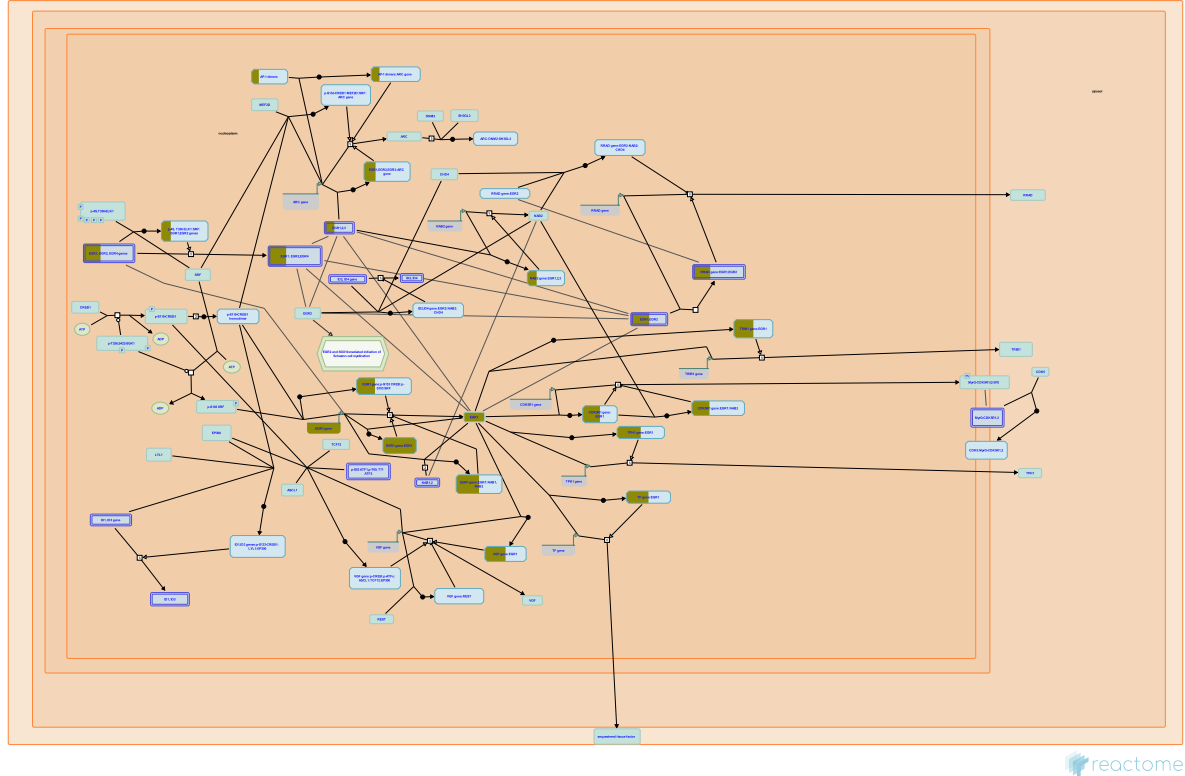
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DNAJA4	Q8WW22	FOS	P01100
HSPA1A	P0DMV8	HSPA6	P17066, P48741

Input	Ensembl Id	Input	Ensembl Id
HSPA1A	ENSG00000204389, ENSG00000215328, ENSG00000234475, ENSG00000235941, ENSG00000237724	HSPA6	ENSG00000173110

Interactors found in this pathway (3)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
EGR1	P18146	Q8N726	FOS	Q6FG41, P01100	P05412, P35638, Q96L91, P18848, P15336, P54646, P01100, Q8N6T3
HSPA1A	P0DMV8	Q13501, Q9NUX5, Q9GZQ8			

8. NGF-stimulated transcription (R-HSA-9031628)



NGF stimulation induces expression of a wide array of transcriptional targets. In rat PC12 cells, a common model for NGF signaling, stimulation with NGF causes cells to exit the cell cycle and undergo a differentiation program leading to neurite outgrowth. This program is driven by the expression of immediate early genes (IEGs), which frequently encode transcription factors regulating the activity of NGF-specific delayed response genes (reviewed in Sheng and Greenberg, 1990; Flavell and Grennberg, 2008; Santiago and Bashaw, 2014).

References

Greenberg ME & Flavell SW (2008). Signaling mechanisms linking neuronal activity to gene expression and plasticity of the nervous system. *Annu. Rev. Neurosci.*, 31, 563-90. [🔗](#)

Santiago C & Bashaw GJ (2014). Transcription factors and effectors that regulate neuronal morphology. *Development*, 141, 4667-80. [🔗](#)

Guérin M, Sheng ZM, Riou G & Andrieu N (1990). Strong association between c-myb and oestrogen-receptor expression in human breast cancer. *Oncogene*, 5, 131-5. [🔗](#)

Edit history

Date	Action	Author
2017-12-01	Created	Rothfels K
2019-08-16	Authored	Rothfels K
2020-01-17	Reviewed	Aletta J M
2020-02-24	Edited	Rothfels K
2023-08-26	Modified	Wright A

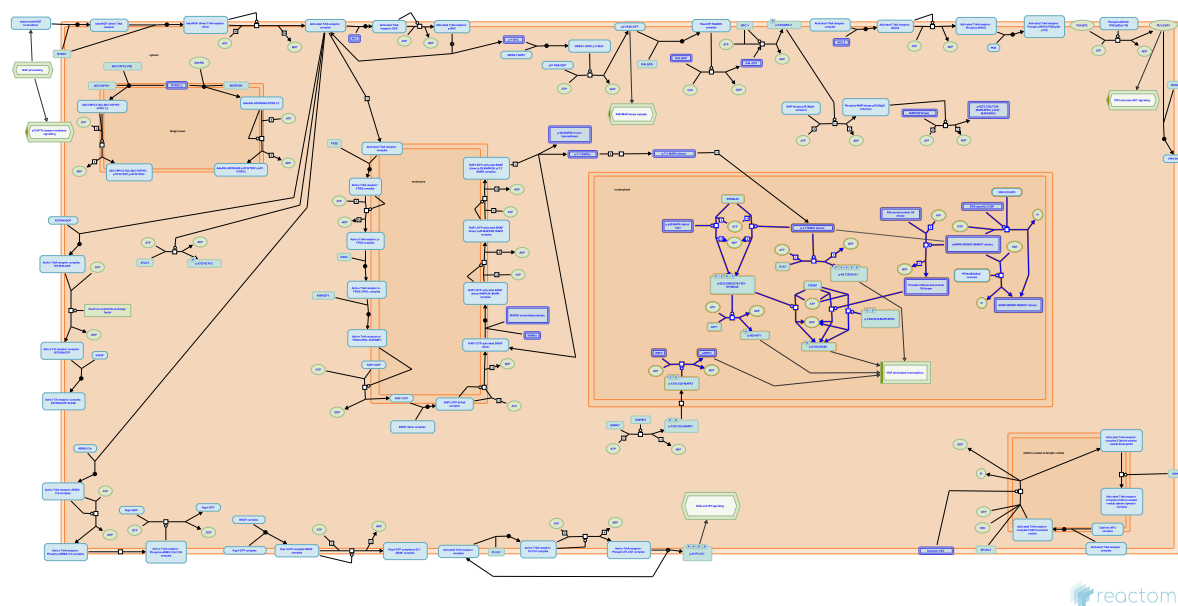
2 submitted entities found in this pathway, mapping to 3 Reactome entities

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Input	Ensembl Id		
EGR1	ENSG00000120738		

Interactors found in this pathway (1)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
FOS	P01100	Q96RU8			

9. Nuclear Events (kinase and transcription factor activation) (R-HSA-198725)



An important function of the kinase cascade triggered by neurotrophins is to induce the phosphorylation and activation of transcription factors in the nucleus to initiate new programs of gene expression. Transcription factors directly activated by neurotrophin signalling are responsible for induction of immediate-early genes, many of which are transcription factors. These in turn are involved in the induction of delayed-early genes.

References

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Edit history

Date	Action	Author
2006-10-10	Authored	Annibali D, Nasi S
2007-07-10	Created	
2007-11-08	Reviewed	Greene LA
2023-08-26	Modified	Wright A

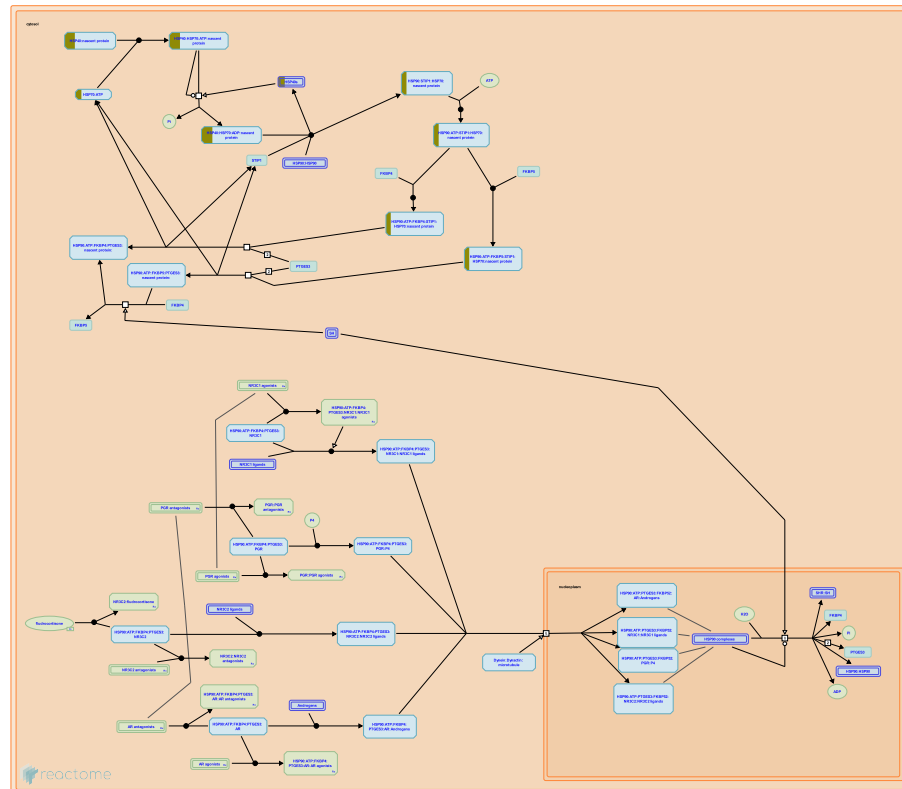
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Input	Ensembl Id		
EGR1	ENSG00000120738		

Interactors found in this pathway (1)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
FOS	P01100	Q96RU8			

10. HSP90 chaperone cycle for steroid hormone receptors (SHR) in the presence of ligand (**R-HSA-3371497**)



Cellular compartments: cytosol.

Steroid hormone receptors (SHR) are transcription factors that become activated upon sensing steroid hormones such as glucocorticoids, mineralocorticoids, progesterone, androgens, or estrogen (Escriva et al 2000; Griekspoor A et al. 2007; Eick GN & Thornton JW. 2011). Depending on SHR type and the presence of ligand, they show different subcellular localizations. Whereas both unliganded and liganded estrogen receptors (ERalpha and ERbeta) are predominantly nuclear, unliganded glucocorticoid (GR) and androgen receptors (AR) are mostly located in the cytoplasm and completely translocate to the nucleus only after binding hormone (Htun H et al. 1999; Stenoien D et al. 2000; Tyagi RK et al. 2000; Cadepond F et al. 1992; Jewell CM et al. 1995; Kumar S et al. 2006). The unliganded mineralocorticoid receptor (MR) is partially cytoplasmic but can be found in nucleus in the ligand-bound or ligand-free form (Nishi M & Kawata M 2007). The progesterone receptor (PR) exists in two forms (PRA and PRB) with different ratios of nuclear versus cytoplasmic localization of the unliganded receptor. In most cell contexts, the PRA isoform is a repressor of the shorter PRB isoform, and without hormone induction it is mostly located in the nucleus, whereas PRB distributes both in the nucleus and in the cytoplasm (Lim CS et al. 1999; Griekspoor A et al. 2007). In the absence of ligand, members of the steroid receptor family remain sequestered in the cytoplasm and/or nucleus in the complex with proteins of HSP70/HSP90 chaperone machinery (Pratt WB & Dittmar KD 1998). The highly dynamic ATP-dependent interactions of SHRs with HSP90 complexes regulate SHR cellular location, protein stability, competency to bind steroid hormones and transcriptional activity (Echeverria PC & Picard D 2010). Understanding the mechanism of ATPase activity of HSP90 is mostly based on structural and functional studies of the *Saccharomyces cerevisiae* Hsp90 complexes (Meyer P et al. 2003, 2004; Ali MM et al. 2006; Prodromou C et al. 2000; Prodromou C 2012). The ATPase cycle of human HSP90 is less well understood, however several studies suggest that the underlying enzymatic mechanisms and a set of conformational changes that accompany the ATPase cycle are highly similar in both species (Richter K et al. 2008; Vaughan CK et al. 2009). Nascent SHR proteins are chaperoned by HSP70 and HSP40 to HSP90 cycle via STIP1 (HOP) (and its TPR domains) (Hernández MP et al. 2002a,b; Echeverria PC & Picard D 2010; Li J et al. 2011). The ATP-bound form of HSP90 leads to the displacement of STIP1 by immunophilins FKBP5 or FKBP4 resulting in conformational changes that allow efficient hormone binding (Li J et al. 2011). PTGES3 (p23) binds to HSP90 complex finally stabilizing it in the conformation with a high hormone binding affinity. After hydrolysis of ATP the hormone bound SHR is released from HSP90 complex. The cytosolic hormone-bound SHR can be transported to the nucleus by several import pathways such as the dynein-based nuclear transport along microtubules involving the transport of the entire HSP90 complex or nuclear localization signals (NLS)-mediated nuclear targeting by importins (Tyagi RK et al. 2000; Cadepond F et al. 1992; Jewell CM et al. 1995; Kumar S et al. 2006). It is worth noting that GR-importin interactions can be ligand-dependent or independent (Freedman & Yamamoto 2004; Picard & Yamamoto 1987). In the nucleus ligand-activated SHR dimerizes, binds specific sequences in the DNA, called Hormone Responsive Elements (HRE), and recruits a number of coregulators that facilitate gene transcription. Nuclear localization is essential for SHRs to transactivate their target genes, but the same receptors also possess non-genomic functions in the cytoplasm.

The Reactome module describes the ATPase-driven conformational cycle of HSP90 that regulates ligand-dependent activation of SHRs.

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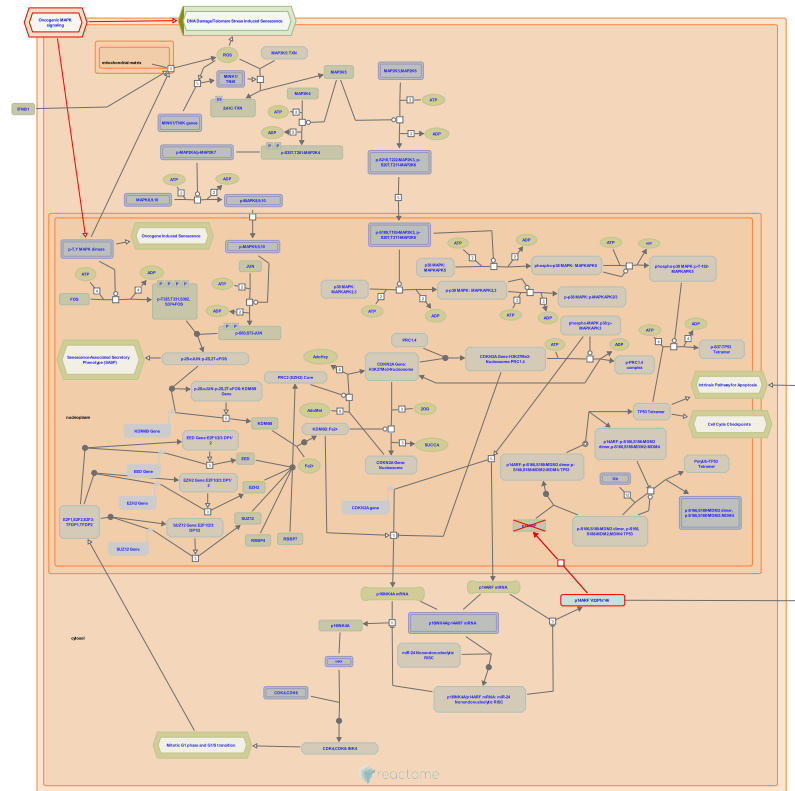
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Date	Action	Author
2013-05-13	Created	Shamovsky V
2016-09-17	Reviewed	Rothfels K
2016-11-19	Authored	Shamovsky V
2017-02-22	Reviewed	Picard D, Echeverria PC
2017-02-25	Edited	Shamovsky V
2023-08-26	Modified	Wright A

2 submitted entities found in this pathway, mapping to 2 Reactome entities

Input	UniProt Id	Input	UniProt Id
DNAJA4	Q8WW22	HSPA1A	P0DMV8

11. Evasion of Oxidative Stress Induced Senescence Due to p14ARF Defects (R-HSA-9646304)



Diseases: cancer.

One of the two main protein products of the CDKN2A gene, p14ARF (CDKN2A transcript 4, CDKN2A-4, ARF), contributes to oxidative stress induced cellular senescence by stabilizing TP53 (p53). The function of p14ARF in p53 stabilization through sequestration of MDM2, a p53 ubiquitin ligase, depends on the nuclear localization of p14ARF and its ability to interact with MDM2. The nuclear localization signal and the MDM2 interaction domain map to the first 15 amino acids of the N-terminus of p14ARF. This region is encoded by the p14ARF-specific exon 1beta of CDKN2A. An independent MDM2-binding domain is localized at the C-terminus of p14ARF (Lohrum et al. 2000). Insertion of 16 nucleotides in exon 1beta results in a frameshift truncation of p14ARF, responsible for a familial melanoma syndrome in which the p16INK4A product of the CDKN2A gene is unaffected. This mutation is rare and has so far been reported in one family only. The mutant protein, p14ARF V22Pfs*46 has the nucleotide localization signal and the N-terminal MDM2 interaction region preserved, but is unable to translocate from the cytosol to the nucleus, possibly due to aberrant conformation (Rizos, Puig et al. 2001), and also lacks the C-terminal MDM2 interaction region. Relocation of wild type p14ARF to the cytosol has been observed in melanoma (Rizos, Darmanian et al. 2001) and aggressive thyroid papillary carcinoma (Ferru et al. 2006). Genomic deletion of exon 1beta, with exons 1alpha, 2 and 3 intact, has been reported in about 30% of melanoma cases with genomic deletions involving the CDKN2A locus (Freedberg et al. 2008). Several different familial melanoma germline mutations map to the exon 1beta splice donor site (Harland et al. 2005).

The ability of p14ARF to localize to the nucleolus also plays a role in p14ARF-mediated stabilization of p53. Mutations in exon 2 of the CDKN2A gene can lead to missense mutations in p14ARF that affect its nucleolar localization and p53 stabilization, but the exact mechanism has not been fully elucidated (Zhang and Xiong 1999, reviewed by Fontana et al. 2019).

References

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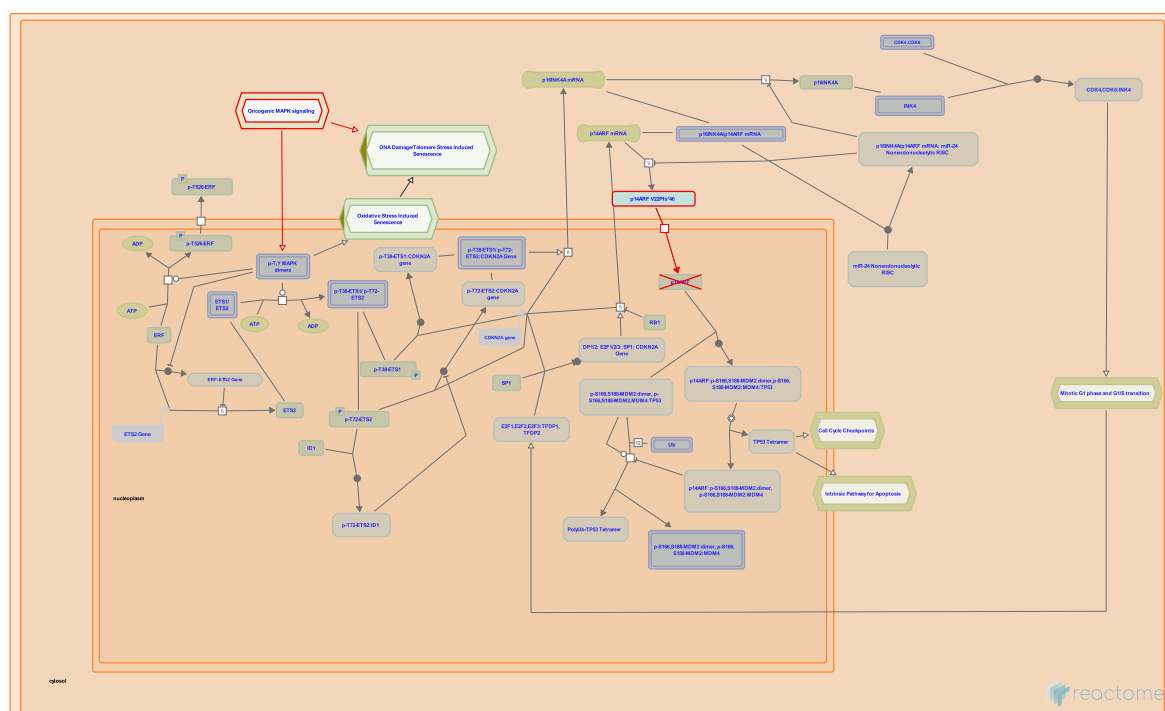
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2019-06-28	Authored	Orlic-Milacic M
2019-07-08	Reviewed	Rizos H
2019-07-16	Edited	Orlic-Milacic M
2019-08-12	Reviewed	Bennett DC
2019-08-14	Edited	Orlic-Milacic M
2023-08-31	Modified	Wright A

Interactors found in this pathway (1)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
EGR1	P18146	Q8N726			

12. Evasion of Oncogene Induced Senescence Due to p14ARF Defects (R-HSA-9646303)



Diseases: cancer.

In cell culture, p14ARF (CDKN2A transcript 4, CDKN2A-4, ARF), one of the two main protein products of the CDKN2A gene, contributes to oncogene induced senescence by stabilizing TP53 (p53). The function of p14ARF in p53 stabilization through sequestration of MDM2, a p53 ubiquitin ligase, depends on the nuclear localization of p14ARF and its ability to interact with MDM2. The nuclear localization signal and the MDM2 interaction domain map to the first 15 amino acids of the N-terminus of p14ARF. This region is encoded by the p14ARF-specific exon 1beta of CDKN2A. An independent MDM2-binding domain localized to the C-terminus of p14ARF (Lohrum et al. 2000). Insertion of 16 nucleotides in exon 1beta results in a frameshift truncation of p14ARF, responsible for a familial melanoma syndrome in which the p16INK4A product of the CDKN2A gene is unaffected. This mutation is rare and has so far been reported in one family only. The mutant protein, p14ARF V22Pfs*46 has the nucleotide localization signal and the N-terminal MDM2 interaction region preserved, but is unable to translocate from the cytosol to the nucleus, possibly due to aberrant conformation (Rizos, Puig et al. 2001), and also lacks the C-terminal MDM2 interaction region. Relocation of wild type p14ARF to the cytosol has been observed in melanoma (Rizos, Darmanian et al. 2001) and aggressive thyroid papillary carcinoma (Ferru et al. 2006). Genomic deletion of exon 1beta, with exons 1alpha, 2 and 3 intact, has been reported in about 30% of melanoma cases with genomic deletions involving the CDKN2A locus (Freedberg et al. 2008). Several different familial melanoma germline mutations map to the exon 1beta splice donor site (Harland et al. 2005).

The ability of p14ARF to localize to the nucleolus also plays a role in p14ARF-mediated stabilization of p53. Mutations in exon 2 of the CDKN2A gene can lead to missense mutations in p14ARF that affect its nucleolar localization and p53 stabilization, but the exact mechanism has not been fully elucidated (Zhang and Xiong 1999, reviewed by Fontana et al. 2019).

References

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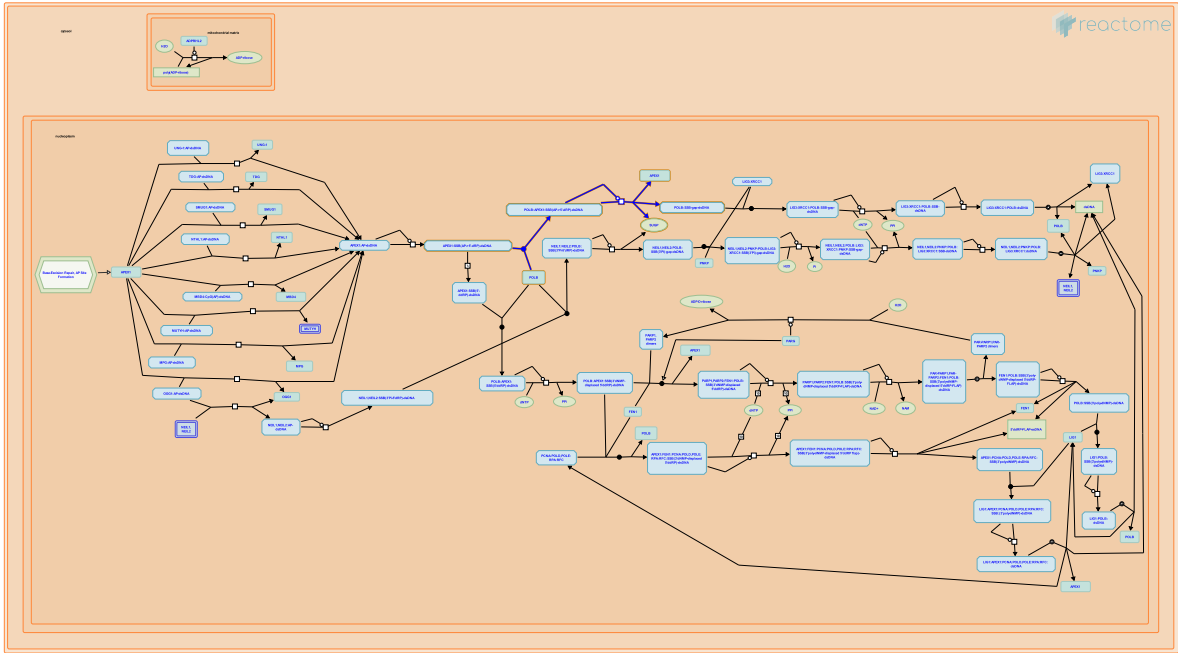
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2019-07-08	Reviewed	Rizos H
2019-07-16	Edited	Orlic-Milacic M
2019-08-12	Reviewed	Bennett DC
2019-08-14	Edited	Orlic-Milacic M
2023-08-31	Modified	Wright A

Interactors found in this pathway (1)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
EGR1	P18146	Q8N726			

13. Abasic sugar-phosphate removal via the single-nucleotide replacement pathway (R-HSA-73930)



Cellular compartments: nucleoplasm.

Abasic sugar phosphate removal via the single nucleotide replacement pathway requires displacement of DNA glycosylase by APEX1, APEX1-mediated endonucleolytic cleavage at the 5' side of the base free deoxyribose residue, recruitment of POLB to the AP site and excision of the abasic sugar phosphate (5'dRP) residue at the strand break (Lindahl and Wood, 1999).

References

Lindahl T & Wood RD (1999). Quality control by DNA repair. Science, 286, 1897-905. [↗](#)

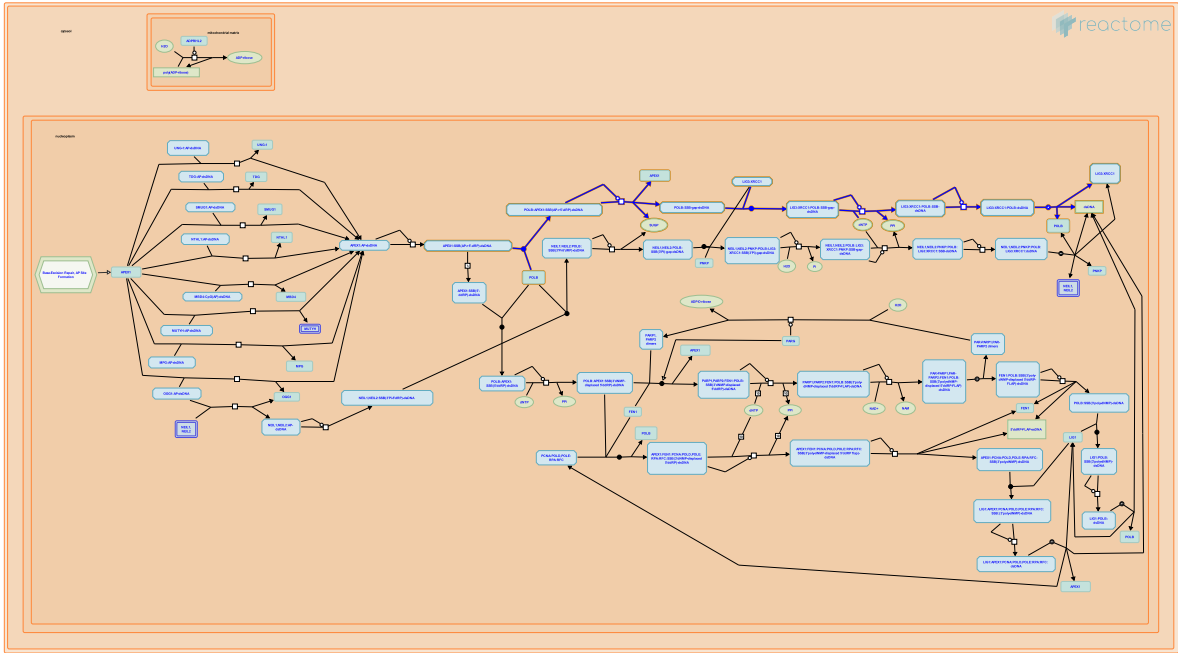
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Date	Action	Author
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2004-02-03	Edited	Matthews L
2004-02-03	Authored	Matthews L
2014-12-04	Revised	Orlic-Milacic M
2014-12-04	Edited	Orlic-Milacic M
2014-12-22	Reviewed	Borowiec JA
2023-08-26	Modified	Wright A

Interactors found in this pathway (1)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
HSPA1A	P08107	P27695			

14. Resolution of AP sites via the single-nucleotide replacement pathway (R-HSA-110381)



Cellular compartments: nucleoplasm.

The single nucleotide replacement pathway of base excision repair appears to facilitate the repair of most damaged bases. Following DNA glycosylase mediated cleavage of the damaged base, the endonuclease APEX1 is recruited to the site of damage where it cleaves the 5' side of the abasic (AP) deoxyribose residue. DNA polymerase beta (POLB) then cleaves the 3' side of the AP sugar phosphate, thus excising the AP residue. APEX1 is subsequently released, the XRCC1:LIG3 complex is recruited, and POLB mediates the synthesis of the replacement residue. Following LIG3 mediated ligation of the replaced residue, the XRCC1:LIG3 complex dissociates from DNA (Lindahl and Wood, 1999). An alternative BER pathway is employed when the structure of the terminal sugar phosphate is such that it cannot be cleaved by the AP lyase activity of POLB.

References

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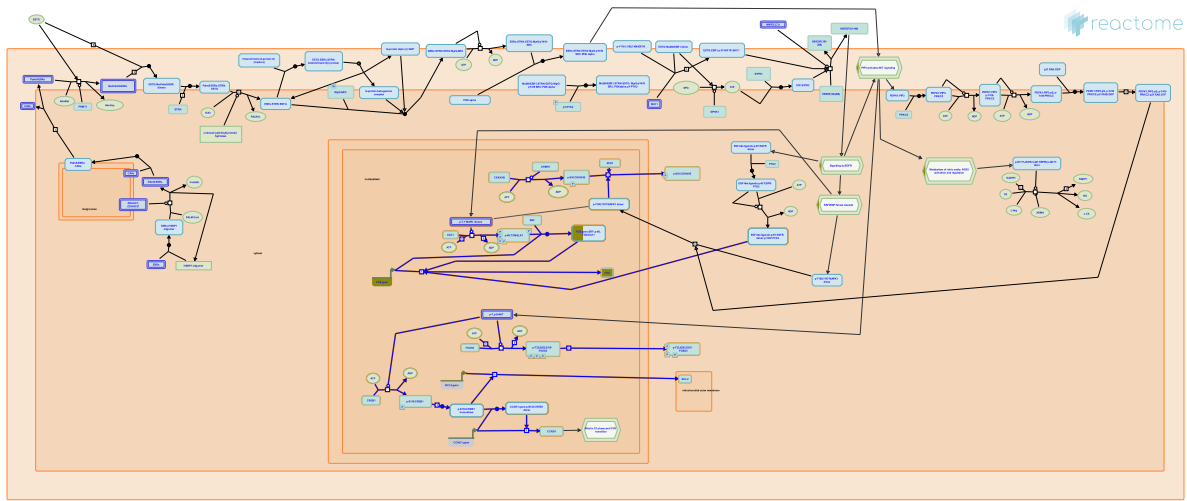
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2014-12-04	Revised	Orlic-Milacic M
2014-12-04	Edited	Orlic-Milacic M
2014-12-22	Reviewed	Borowiec JA
2023-08-26	Modified	Wright A

Interactors found in this pathway (1)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
HSPA1A	P08107	P27695			

15. Estrogen-dependent nuclear events downstream of ESR-membrane signaling
(R-HSA-9634638)



Although membrane-localized estrogen receptors stimulate rapid, transcription-independent responses such as calcium mobilization and alterations to the fibronectin matrix to affect cell migration, among others, the pathways activated by rapid signaling may also ultimately affect nuclear events. Activation of MAPK and PI3K/AKT pathways downstream of membrane-localized ESR1 contributes to estrogen-responsive changes in cellular proliferation and survival in part through changes in gene expression (reviewed in Levin et al, 2005; Lange et al, 2007; Le Romancer et al, 2011).

References

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Edit history

Date	Action	Author
2018-12-15	Authored	Rothfels K
2019-01-12	Created	Rothfels K
2019-02-20	Reviewed	Levin ER
2023-08-26	Modified	Wright A

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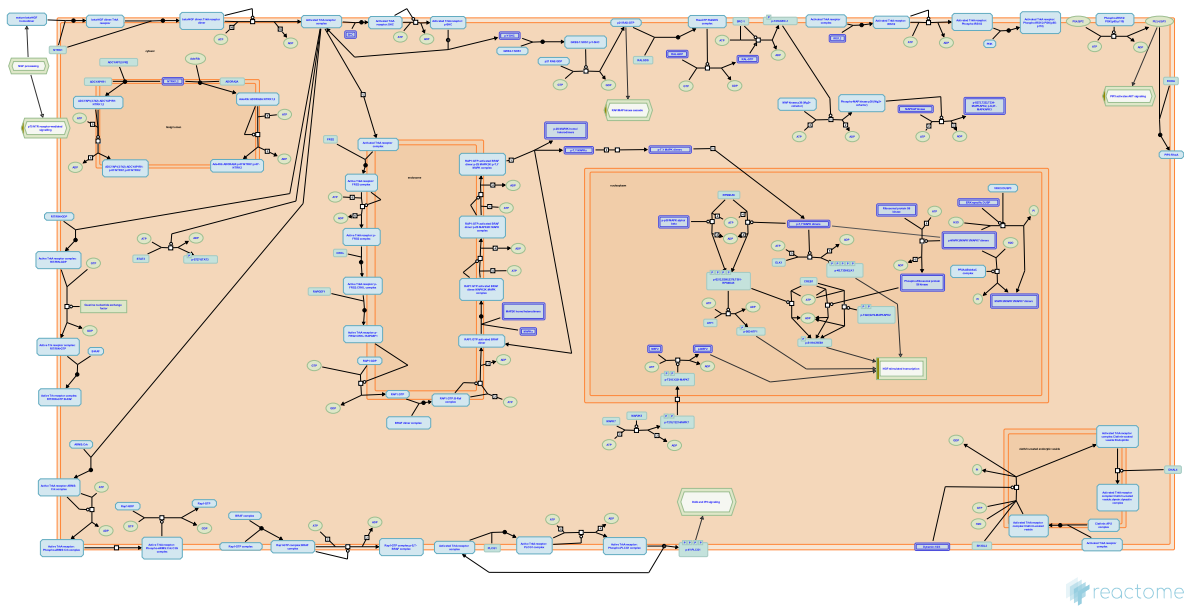
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FOS	P01100

Input	Ensembl Id
FOS	ENSG00000170345

Interactors found in this pathway (1)

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FOS	P01100	P01100			

16. Signaling by NTRK1 (TRKA) (R-HSA-187037)



Trk receptors signal from the plasma membrane and from intracellular membranes, particularly from early endosomes. Signalling from the plasma membrane is fast but transient; signalling from endosomes is slower but long lasting. Signalling from the plasma membrane is annotated here. TRK signalling leads to proliferation in some cell types and neuronal differentiation in others. Proliferation is the likely outcome of short term signalling, as observed following stimulation of EGFR (EGF receptor). Long term signalling via TRK receptors, instead, was clearly shown to be required for neuronal differentiation in response to neurotrophins.

References

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Date	Action	Author
2006-09-07	Created	Jassal B
2006-10-10	Edited	Jassal B
2006-10-10	Authored	Annibali D, Nasi S
2007-11-08	Reviewed	Greene LA
2023-08-26	Modified	Wright A

2 submitted entities found in this pathway, mapping to 3 Reactome entities

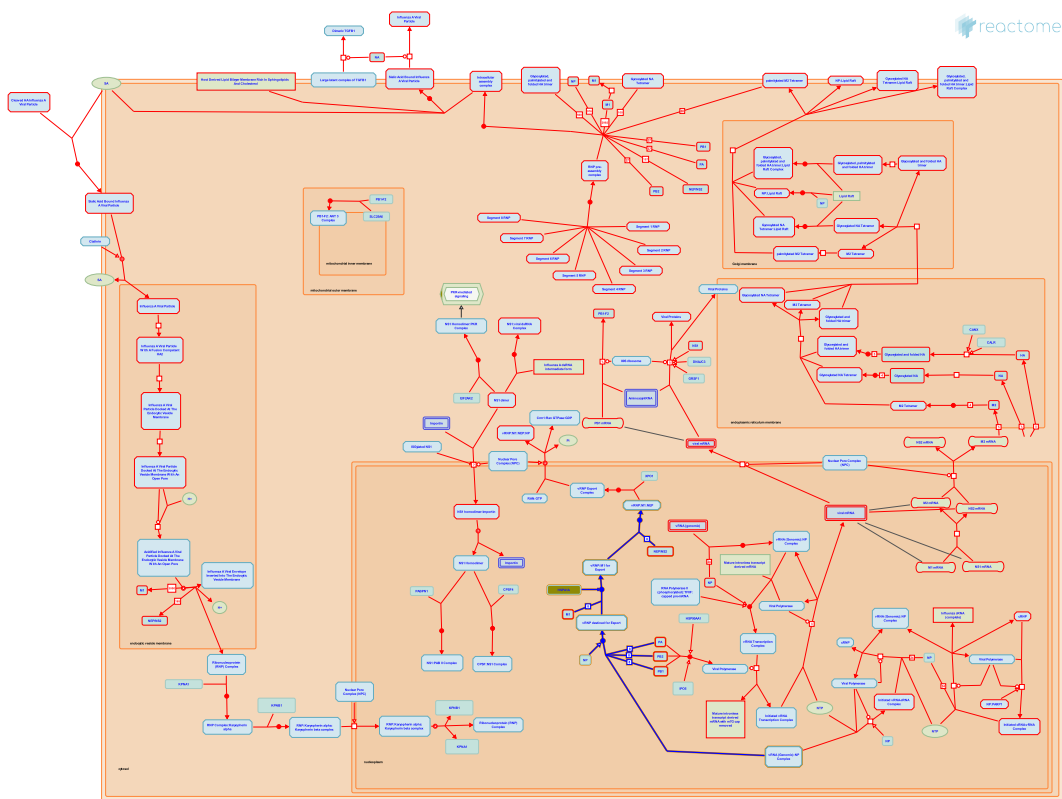
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Input	Ensembl Id
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Interactors found in this pathway (1)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
FOS	P01100	Q96RU8			

17. Viral RNP Complexes in the Host Cell Nucleus (R-HSA-168330)



Cellular compartments: nucleoplasm.

Diseases: influenza.

Viral RNP is assembled in the host cell nucleus through the interaction of full-length negative strand viral RNA (vRNA) and the viral nucleocapsid (NP) and matrix (M1) proteins. Studies of interactions of the purified components in vitro and of tissue culture model systems expressing various combinations of the components have established roles for both NP and M1 proteins in the assembly of a complex that has the physical properties of vRNP purified from virions and that can be exported from the host cell nucleus (Whittaker, 1996; Huang, 2001; Baudin, 2001). Viral RNP complexes have been found in the nucleoplasm, and also in the nuclear periphery, associated with the nuclear matrix or chromatin, particularly for vRNA-containing complexes and M1 protein (Elton, 2005; Garcia-Robles, 2005; Takizawa et al., 2006).

References

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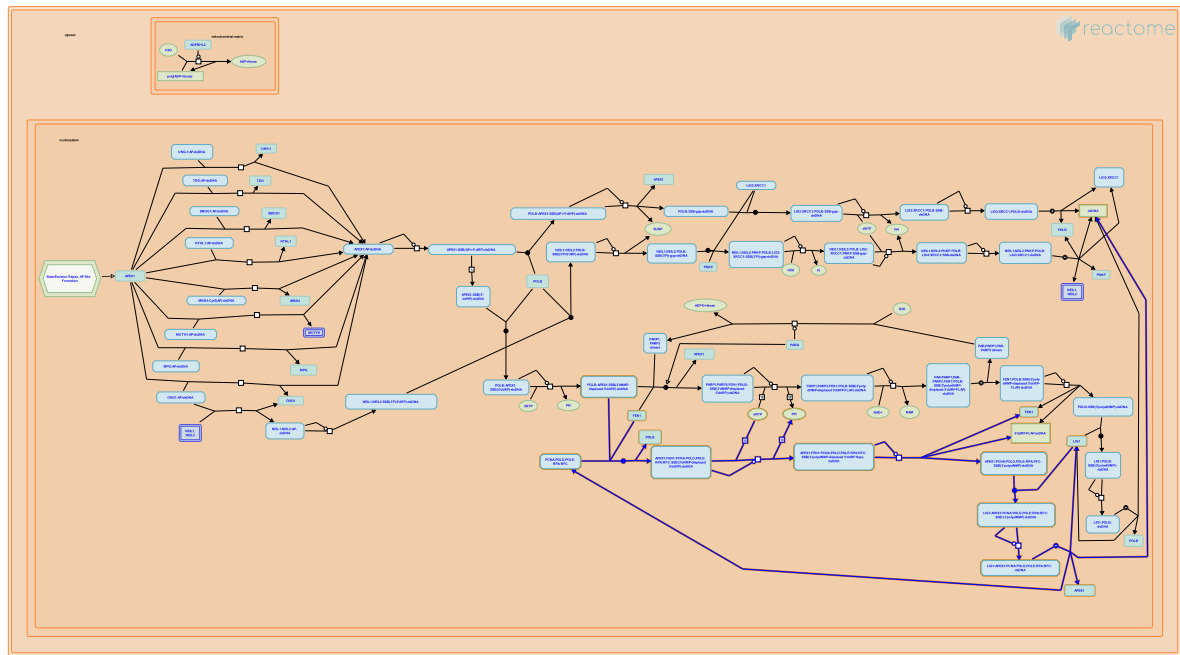
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Date	Action	Author
2007-02-13	Reviewed	Squires B
2007-02-13	Authored	Garcia-Sastre A, Bortz E
2007-04-17	Created	
2023-03-08	Modified	Matthews L

1 submitted entities found in this pathway, mapping to 1 Reactome entities

Input	UniProt Id
HSPA1A	P0DMV8

18. PCNA-Dependent Long Patch Base Excision Repair (R-HSA-5651801)



Cellular compartments: nucleoplasm.

Long-patch base excision repair (BER) can proceed through PCNA-dependent DNA strand displacement synthesis by replicative DNA polymerases - DNA polymerase delta complex (POLD) or DNA polymerase epsilon (POLE) complex. The PCNA-dependent branch of long-patch BER may occur in cells in the S phase of the cell cycle, when the replication complexes that contain PCNA, POLD or POLE, RPA and RFC are available. POLB incorporates the first nucleotide at the 3'-end of APEX1-generated single strand break (SSB), thus displacing the damaged AP (abasic) dideoxyribose phosphate residue at the 5'-end of SSB (5'ddRP). PCNA is recruited to BER sites by APEX1 and flap endonuclease FEN1, and loaded onto damaged DNA by RFC. POLD and POLE in complex with PCNA continue the displacement DNA strand synthesis. FEN1 cleaves the displaced DNA strand with the AP residue (5'ddRP), and DNA ligase I (LIG1) ligates the multiple nucleotide patch at the 3' end of the SSB with the FEN1-processed 5'-end of the SSB (Klungland and Lindahl 1997, Stucki et al. 1998, Dianov et al. 1999, Matsumoto et al. 1999, Podlutzky et al. 2001, Dianova et al. 2001, Ranalli et al. 2002).

References

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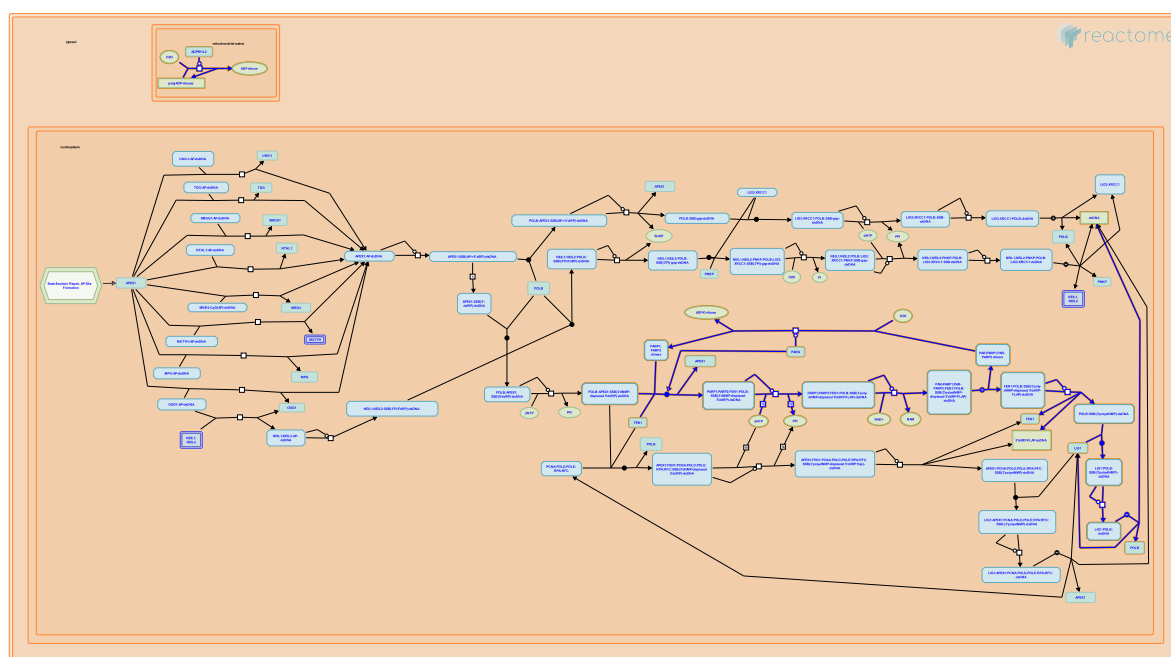
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2014-12-04	Authored	Orlic-Milacic M
2014-12-22	Reviewed	Borowiec JA
2023-08-26	Modified	Wright A

Interactors found in this pathway (1)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
HSPA1A	P08107	P27695			

19. POLB-Dependent Long Patch Base Excision Repair (R-HSA-110362)



Cellular compartments: nucleoplasm.

During POLB-dependent long patch base excision repair (BER), PARP1 and/or PARP2 is recruited to the BER site along with flap endonuclease FEN1. PARP1 and/or PARP2 and FEN1 facilitate POLB-mediated strand displacement synthesis which involves incorporation of 2-10 nucleotides at the 3' end of the APEX1-created single strand break (SSB). After the DNA strand displacement synthesis is completed and the displaced strand is cleaved, POLB recruits DNA ligase I (LIG1) to ligate the SSB (Klungland and Lindahl 1997, Dimitriadis et al. 1998, Prasad et al. 2001, Lavrik et al. 2001, Tomkinson et al. 2001, Ranalli et al. 2002, Cistulli et al. 2004, Liu et al. 2005).

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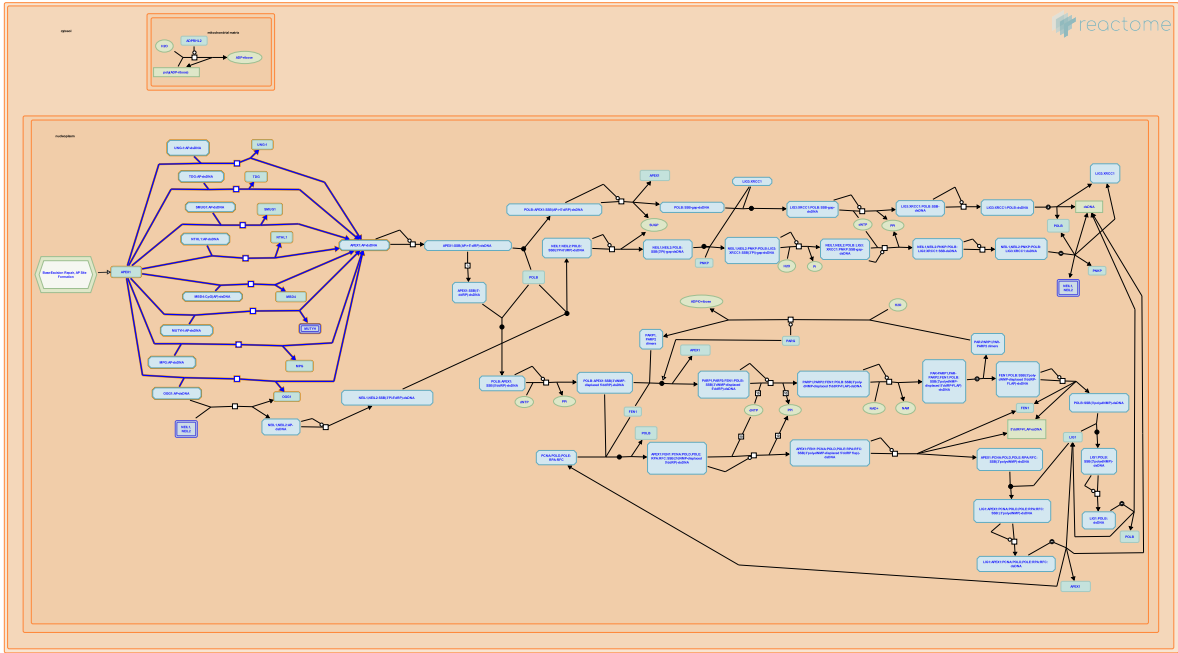
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Date	Action	Author
2004-01-29	Created	Matthews L
2004-02-03	Edited	Matthews L
2004-02-03	Authored	Matthews L
2014-12-04	Revised	Orlic-Milacic M
2014-12-04	Edited	Orlic-Milacic M
2014-12-22	Reviewed	Borowiec JA
2023-08-26	Modified	Wright A

Interactors found in this pathway (1)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
HSPA1A	P08107	P27695			

20. Displacement of DNA glycosylase by APEX1 (R-HSA-110357)



Cellular compartments: nucleoplasm.

Following cleavage of the damaged base, DNA glycosylase is displaced by APEX1, an AP endonuclease (Parikh et al. 1998).

References

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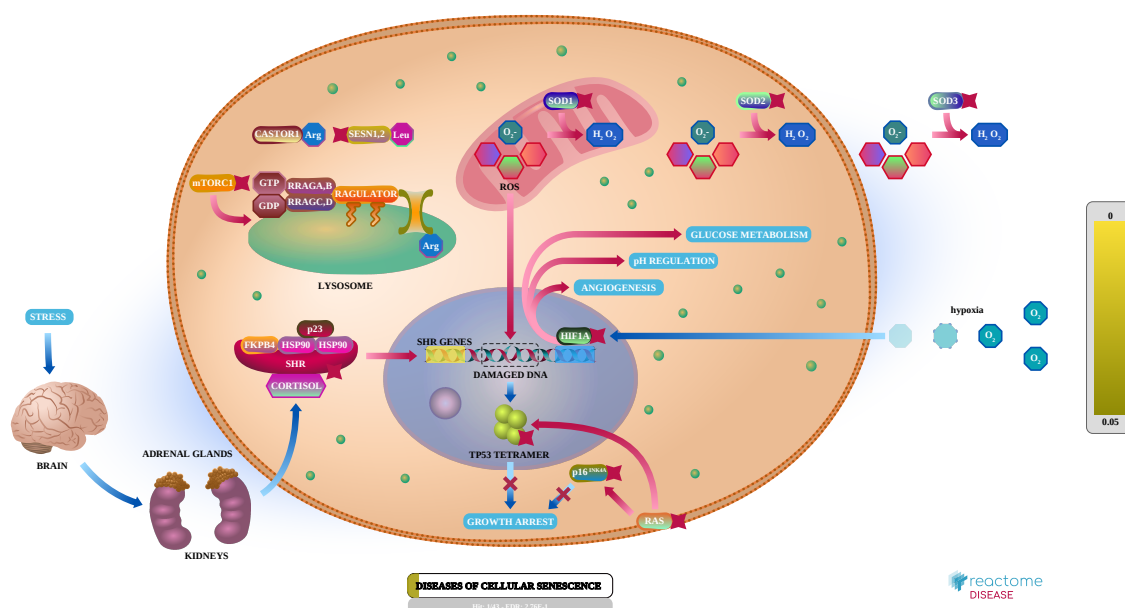
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2004-02-03	Authored	Matthews L
2014-12-04	Revised	Orlic-Milacic M
2014-12-04	Edited	Orlic-Milacic M
2014-12-22	Reviewed	Borowiec JA
2023-08-26	Modified	Wright A

Interactors found in this pathway (1)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
HSPA1A	P08107	P27695			

21. Diseases of cellular response to stress (R-HSA-9675132)



Cells are subject to external and internal stressors, such as foreign molecules that perturb metabolic or signaling processes, cellular respiration-generated reactive oxygen species that can cause DNA damage, oxygen and nutrient deprivation, and changes in temperature or pH. The ability of cells and tissues to respond to stress is essential to the maintenance of tissue homeostasis (Kültz 2005) and dysregulation of cellular response to stress is involved in disease.

So far, we have captured diseases of cellular senescence.

Impaired cellular senescence contributes to malignant transformation and cancer development by enabling continuous proliferation of damaged cells. On the other hand, presence of an excessive number of senescent cells that are not cleared by the immune system promotes tissue inflammation and creates a microenvironment suitable for growth of neighboring malignant cells. In addition to cancer, senescence is also involved in other age-related diseases such as atherosclerosis, osteoarthritis, chronic obstructive lung disease, and diabetes (Childs et al. 2015, He and Sharpless 2017, Hamsanathan et al. 2019, Faget et al. 2019, Gorgoulis et al. 2019, Rhinn et al. 2019). Senotherapy is a new field of pharmacology that aims to therapeutically target senescence to improve healthy aging and age-related diseases (Schmitt 2017, Gorgoulis et al. 2019).

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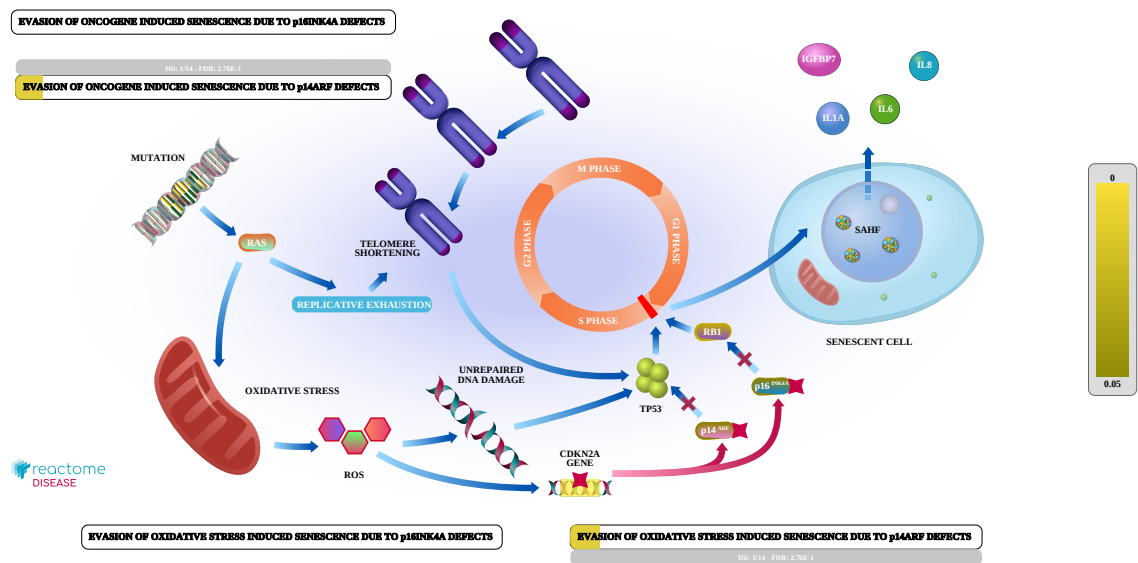
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2020-02-21	Authored	Orlic-Milacic M
2020-02-24	Edited	Orlic-Milacic M
2020-02-24	Reviewed	D'Eustachio P
2023-03-08	Modified	Matthews L

Interactors found in this pathway (1)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
EGR1	P18146	Q8N726			

22. Diseases of Cellular Senescence (R-HSA-9630747)



Diseases: cancer.

Cellular senescence plays an important role in normal aging, as well as in age-related diseases. Impaired cellular senescence contributes to malignant transformation and cancer development. Presence of an excessive number of senescent cells that are not cleared by the immune system, however, promotes tissue inflammation and creates a microenvironment suitable for growth of neighboring malignant cells. Besides cancer, senescence is also involved in atherosclerosis, osteoarthritis and diabetes (Childs et al. 2015, He and Sharpless 2017).

Evasion of oncogene-induced senescence, at least in cell culture, can occur due to loss-of-function (LOF) mutation in the CDKN2A gene product p16INK4A that acts as a cyclin-dependent kinase inhibitor (reviewed in Sharpless and Sherr 2015). LOF mutations in the CDKN2A gene that affect its other protein product, p14ARF, involved in stabilization of TP53 protein (p53), can contribute to evasion of oncogene-induced senescence (reviewed in Fontana et al. 2019).

LOF mutations in p16INK4A and p14ARF also contribute to evasion of oxidative stress-induced senescence (reviewed in Sharpless and Sherr 2015, and Fontana et al. 2019, respectively).

References

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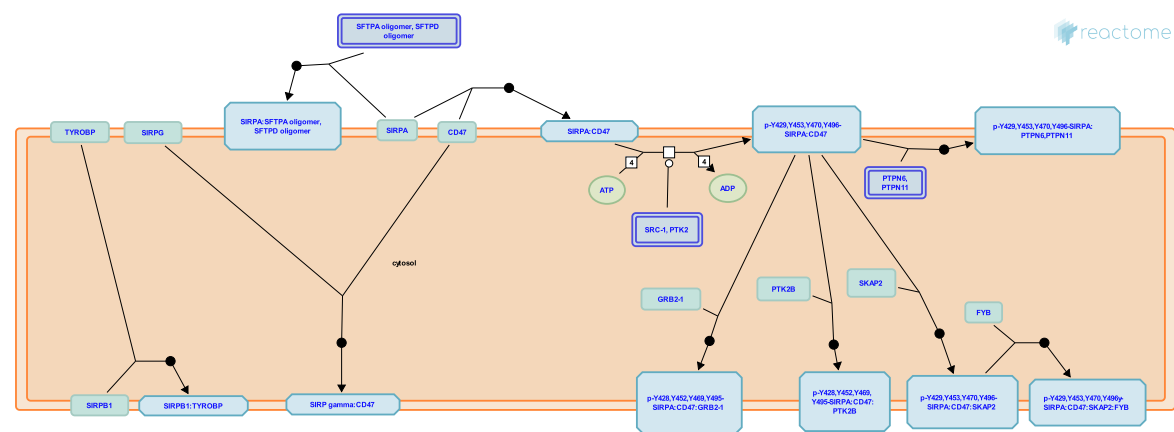
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2018-12-24	Authored	Orlic-Milacic M

Date	Action	Author
2019-04-23	Reviewed	Bennett DC
2019-05-07	Edited	Orlic-Milacic M
2019-06-03	Reviewed	Nathan V, Hayward NK
2023-03-08	Modified	Matthews L

Interactors found in this pathway (1)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
EGR1	P18146	Q8N726			

23. Signal regulatory protein family interactions (R-HSA-391160)



Cellular compartments: plasma membrane.

Signal regulatory protein alpha (SIRPA, SHPS1, CD172a) is a transmembrane protein expressed mostly on myeloid cells. CD47, a widely expressed transmembrane protein, is a ligand for SIRP alpha, with the two proteins constituting a cell-cell communication system. The interaction of SIRPA with CD47 is important for the regulation of migration and phagocytosis. SIRPA functions as a docking protein to recruit and activate PTPN6 (SHP-1) or PTPN11 (SHP-2) at the cell membrane in response to extracellular stimuli. SIRPA also binds other intracellular proteins including the adaptor molecules Src kinase-associated protein (SKAP2 SKAP55hom/R), Fyn-binding protein/SLP-76-associated phosphoprotein (FYB/SLAP-130) and the tyrosine kinase PYK2. SIRPA also binds the extracellular proteins, surfactant-A (SP-A) and surfactant-D (SP-D).

The SIRP family members SIRPB and SIRPG show high sequence similarity and similar extracellular structural topology, including three Ig domains, but their ligand binding topology might differ. SIRPB is expressed on myeloid cells, including monocytes, granulocytes and DCs. It has no known natural ligand. SIRPG can bind CD47 but with lower affinity than SIRPA.

References

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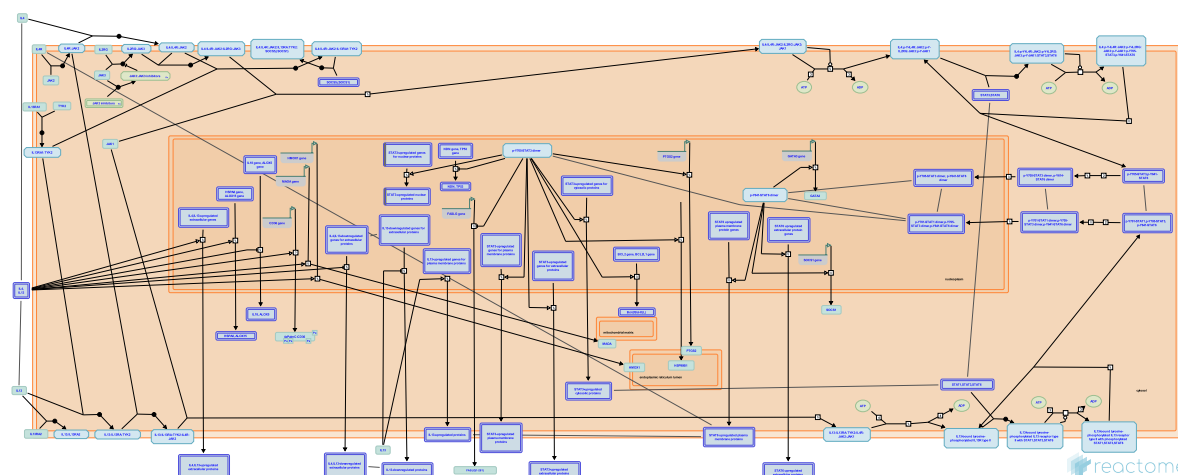
Date	Action	Author
2009-02-12	Edited	Garapati P V

Date	Action	Author
2009-02-12	Authored	Garapati P V
2009-02-12	Created	Garapati P V
2010-05-20	Reviewed	Barclay AN
2023-08-26	Modified	Wright A

Interactors found in this pathway (1)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
FOS	P01100	Q9P1W8			

24. Interleukin-4 and Interleukin-13 signaling (R-HSA-6785807)



Interleukin-4 (IL4) is a principal regulatory cytokine during the immune response, crucially important in allergy and asthma (Nelms et al. 1999). When resting T cells are antigen-activated and expand in response to Interleukin-2 (IL2), they can differentiate as Type 1 (Th1) or Type 2 (Th2) T helper cells. The outcome is influenced by IL4. Th2 cells secrete IL4, which both stimulates Th2 in an autocrine fashion and acts as a potent B cell growth factor to promote humoral immunity (Nelms et al. 1999).

Interleukin-13 (IL13) is an immunoregulatory cytokine secreted predominantly by activated Th2 cells. It is a key mediator in the pathogenesis of allergic inflammation. IL13 shares many functional properties with IL4, stemming from the fact that they share a common receptor subunit. IL13 receptors are expressed on human B cells, basophils, eosinophils, mast cells, endothelial cells, fibroblasts, monocytes, macrophages, respiratory epithelial cells, and smooth muscle cells, but unlike IL4, not T cells. Thus IL13 does not appear to be important in the initial differentiation of CD4 T cells into Th2 cells, rather it is important in the effector phase of allergic inflammation (Hershey et al. 2003).

IL4 and IL13 induce “alternative activation” of macrophages, inducing an anti-inflammatory phenotype by signaling through IL4R alpha in a STAT6 dependent manner. This signaling plays an important role in the Th2 response, mediating anti-parasitic effects and aiding wound healing (Gordon & Martinez 2010, Loke et al. 2002)

There are two types of IL4 receptor complex (Andrews et al. 2006). Type I IL4R (IL4R1) is predominantly expressed on the surface of hematopoietic cells and consists of IL4R and IL2RG, the common gamma chain. Type II IL4R (IL4R2) is predominantly expressed on the surface of nonhematopoietic cells, it consists of IL4R and IL13RA1 and is also the type II receptor for IL13. (Obiri et al. 1995, Aman et al. 1996, Hilton et al. 1996, Miloux et al. 1997, Zhang et al. 1997). The second receptor for IL13 consists of IL4R and Interleukin-13 receptor alpha 2 (IL13RA2), sometimes called Interleukin-13 binding protein (IL13BP). It has a high affinity receptor for IL13 (Kd = 250 pmol/L) but is not sufficient to render cells responsive to IL13, even in the presence of IL4R (Donaldson et al. 1998). It is reported to exist in soluble form (Zhang et al. 1997) and when overexpressed reduces JAK-STAT signaling (Kawakami et al. 2001). It's function may be to prevent IL13 signalling via the functional IL4R:IL13RA1 receptor. IL13RA2 is overexpressed and enhances cell invasion in some human cancers (Joshi & Puri 2012).

The first step in the formation of IL4R1 (IL4:IL4R:IL2RB) is the binding of IL4 with IL4R (Hoffman et al. 1995, Shen et al. 1996, Hage et al. 1999). This is also the first step in formation of IL4R2 (IL4:IL4R:IL13RA1). After the initial binding of IL4 and IL4R, IL2RB binds (LaPorte et al. 2008), to form IL4R1. Alternatively, IL13RA1 binds, forming IL4R2. In contrast, the type II IL13 complex (IL13R2) forms with IL13 first binding to IL13RA1 followed by recruitment of IL4R (Wang et al. 2009).

Crystal structures of the IL4:IL4R:IL2RG, IL4:IL4R:IL13RA1 and IL13:IL4R:IL13RA1 complexes have been determined (LaPorte et al. 2008). Consistent with these structures, in monocytes IL4R is tyrosine phosphorylated in response to both IL4 and IL13 (Roy et al. 2002, Gordon & Martinez 2010) while IL13RA1 phosphorylation is induced only by IL13 (Roy et al. 2002, LaPorte et al. 2008) and IL2RG phosphorylation is induced only by IL4 (Roy et al. 2002).

Both IL4 receptor complexes signal through Jak/STAT cascades. IL4R is constitutively-associated with JAK2 (Roy et al. 2002) and associates with JAK1 following binding of IL4 (Yin et al. 1994) or IL13 (Roy et al. 2002). IL2RG constitutively associates with JAK3 (Boussiotis et al. 1994, Russell et al. 1994). IL13RA1 constitutively associates with TYK2 (Umeshita-Suyama et al. 2000, Roy et al. 2002, LaPorte et al. 2008, Bhattacharjee et al. 2013).

IL4 binding to IL4R1 leads to phosphorylation of JAK1 (but not JAK2) and STAT6 activation (Takeda et al. 1994, Ratthe et al. 2007, Bhattacharjee et al. 2013).

IL13 binding increases activating tyrosine-99 phosphorylation of IL13RA1 but not that of IL2RG. IL4 binding to IL2RG leads to its tyrosine phosphorylation (Roy et al. 2002). IL13 binding to IL4R2 leads to TYK2 and JAK2 (but not JAK1) phosphorylation (Roy & Cathcart 1998, Roy et al. 2002).

Phosphorylated TYK2 binds and phosphorylates STAT6 and possibly STAT1 (Bhattacharjee et al. 2013).

A second mechanism of signal transduction activated by IL4 and IL13 leads to the insulin receptor substrate (IRS) family (Kelly-Welch et al. 2003). IL4R1 associates with insulin receptor substrate 2 and activates the PI3K/Akt and Ras/MEK/Erk pathways involved in cell proliferation, survival and translational control. IL4R2 does not associate with insulin receptor substrate 2 and consequently the PI3K/Akt and Ras/MEK/Erk pathways are not activated (Busch-Dienstfertig & González-Rodríguez 2013).

References

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Date	Action	Author
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2015-07-01	Created	Jupe S
2016-09-02	Edited	Jupe S
2016-09-02	Reviewed	Leibovich SJ

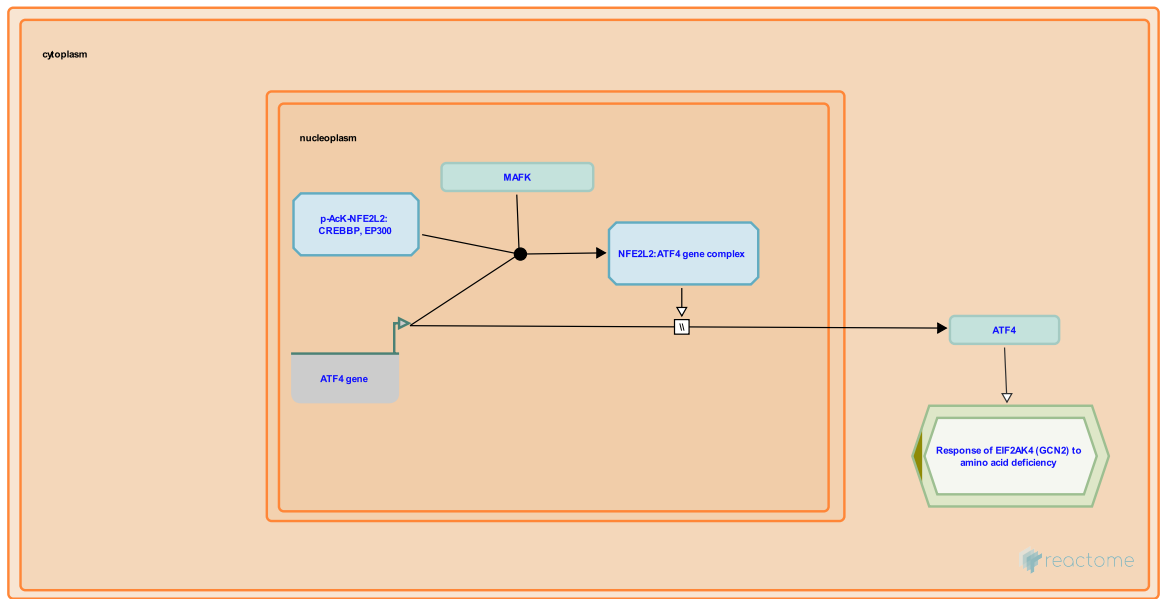
Date	Action	Author
2023-08-31	Modified	Wright A

1 submitted entities found in this pathway, mapping to 2 Reactome entities

Input	UniProt Id
FOS	P01100

Input	Ensembl Id
FOS	ENSG00000170345

25. NFE2L2 regulating ER-stress associated genes (R-HSA-9818035)



Cellular compartments: nucleoplasm.

Subpathway representing ER-stress-associated genes regulated by NFE2L2 (NRF2). Activating transcription factor 4 (ATF4) is a stress-induced transcription factor that is frequently upregulated in cancer cells. ATF4 controls the expression of a wide range of adaptive genes that allow cells to endure periods of stress, such as hypoxia or amino acid limitation. However, under persistent stress conditions, ATF4 promotes the induction of apoptosis (Wortel et al, 2017). ATF4 is also known to regulate serine and glycine biosynthesis in NSCLC (Denicola et al, 2016)

References

Edit history

Date	Action	Author
2022-10-10	Created	Tiwari K
2023-08-08	Edited	Tiwari K
2023-08-08	Authored	Tiwari K
2023-08-21	Modified	Matthews L

Interactors found in this pathway (1)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
FOS	P01100	P18848			

6. Identifiers found

Below is a list of the input identifiers that have been found or mapped to an equivalent element in Reactome, classified by resource.

5 of the submitted entities were found, mapping to 14 Reactome entities

Input	UniProt Id	Input	UniProt Id	Input	UniProt Id
DNAJA4	Q8WW22	EGR1	P18146	FOS	P01100
HSPA1A	P0DMV8	HSPA6	P17066, P48741		

Input	Ensembl Id	Input	Ensembl Id
EGR1	ENSG00000120738	FOS	ENSG00000170345
HSPA1A	ENSG00000204389, ENSG00000215328, ENSG00000234475, ENSG00000235941, ENSG00000237724	HSPA6	ENSG00000173110

Interactors (5)

Input	UniProt Id	Interacts with	Input	UniProt Id	Interacts with
DNAJA4	Q8WW22-2	Q8N9N5	EGR1	P18146	Q8N726
FOS	Q6FG41, P01100	P05412, P35638, Q96L91, P18848, P15336, P54646, P01100, Q8N6T3	HSPA1A	P0DMV8	Q13501, Q9NUX5, Q9GZQ8
HSPA6	P17066	P04049			

7. Identifiers not found

These 1 identifiers were not found neither mapped to any entity in Reactome.

AL627171.2